Topological Hochschild Homology

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Abstract

These notes aim to give an obviously functorial construction of topological Hochschild homology and topological cyclic homology in the category of orthogonal spectra following the relatively recent work of Nikolaus-Scholze. We also give an overview of the old calculations of Hesselholt-Madsen of topological cyclic homology of perfect fields of char $p > 0$.

Contents

1 Introduction

Given an associative $\mathbb{Z}$ algebra A, the topological Hochschild homology, THH(A) is a spectrum refinement of the classical Hochschild homology HH(A) of A. The Hochschild Homology of an associative algebra A is defined as the object $A \otimes^{\mathbb{L}}_{A \otimes_{\mathbb{Z}} A^{op}} A$[1] of the derived category $D(\mathbb{Z})$. One can, via the Dold-Kan correspondence, transport this object to the (homotopy class) of a simplicial abelian group,

$$\dots \;\substack{\longrightarrow\\\longrightarrow\\\longrightarrow\\\longrightarrow}\; A \otimes^{\mathbb{L}}_{\mathbb{Z}} A \otimes^{\mathbb{L}}_{\mathbb{Z}} A \;\substack{\longrightarrow\\\longrightarrow\\\longrightarrow}\; A \otimes^{\mathbb{L}}_{\mathbb{Z}} A \;\substack{\longrightarrow\\\longrightarrow}\; A$$

which we denote by abuse of notation as A.

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[1] See [Wei95, Lem. 9.1.3] to make sense of this.

In fact this simplicial object can be identified with a cyclic set in the sense of Connes, where the action of C_p comes from C_p acting on the n-th degree, $A^{\otimes_{\mathbb{Z}} n}$, of the simplicial object by permuting the factors cyclically. This allows us to consider the geometric realization $|A|$ of A, which is then a bonafide space with an S^1 action and one can identify HH(A) as the space $|A|$.[2] Note that as a simplicial abelian group, $\mathrm{HH}(A)$ has the homotopy type of a generalised Eilenberg-Maclane space $\prod_{n\geq 0} K(\pi_n A, n)$.

One can transport the object HH(A) of $D(\mathbb{Z})$ to the stable module category over $H\mathbb{Z}$, and hence it can be viewed as a generalised Eilenberg-Maclane spectrum. Then there are finer invariants, in a sense we make precise, that one can obtain from $\mathrm{HH}(A)$, namely the cyclic homology $\mathrm{HC}(A)$ and negative cyclic homology $\mathrm{HC}^{-}(A)$, which may, in nice enough categories of spectra, can be defined as the homotopy orbits $\mathrm{HH}(A)_{hS^1}$ and homotopy fixed points $\mathrm{HH}(A)^{hS^1}$ (see, for example, [Hoy15]).

There is a Chern character map between $\mathrm{ch}_i^{-} : \pi_i \mathrm{K}(A) \to \pi_i \mathrm{HC}^{-}(A)$, where $\mathrm{K}(A)$ is the K-theory spectrum of A, see for example [Lod98, Chpt. 8 and 11]. It is so called because in case A is a unital $\mathbb{Q}$-algebra then $\mathrm{HC}^{-}(A)$ is essentially the de-Rham complex of A. Heuristically, if one were to state that algebraic K-theory of a ring studies invertible matrices over that ring upto some symmetry, then the Chern character map may be interpreted as recording the exterior powers of the invertible matrices.

There is always a natural map $\mathrm{HC}_{-}(A) \to \mathrm{HH}(A)$. The Dennis trace map $\mathrm{Dtr} : \pi_i \mathrm{K}(A) \to \pi_i \mathrm{HH}(A)$ is the composition of the natural map $\pi_i \mathrm{HC}^{-} \to \pi_i \mathrm{HH}(A)$.

Goodwillie, in work done in [Goo86], shows that there is a functorial spectra level trace map $\mathrm{tr} : \mathrm{K}(A) \to \mathrm{HC}^{-}(A)$ so that in case $A \to \bar{A}$ is a surjection of associative unital algebras with nilpotent kernel, then the following diagram

$$\begin{array}{ccc} \mathrm{K}(A)_{\mathbb{Q}} & \xrightarrow{\mathrm{tr}} & \mathrm{HC}^{-}(A \otimes \mathbb{Q}) \\ \downarrow & & \downarrow \\ \mathrm{K}(\bar{A})_{\mathbb{Q}} & \xrightarrow{\mathrm{tr}} & \mathrm{HC}^{-}(\bar{A} \otimes \mathbb{Q}) \end{array}$$

is homotopy cartesian. Here $(-)_{\mathbb{Q}}$ is the rationalisation of a spectrum. This trace map is called the Jones-Goodwillie trace map which upon taking homotopy groups agrees with the Dennis trace map defined previously.

One would in general like an integral variant of the above cartesian diagram. It is this effort which resulted in the construction of Topological Hochschild Homology, a spectra version of Hochschild homology, wherein one takes the Hochschild complex over the spehere spectrum $\mathbb{S}$ instead of $\mathbb{Z}$.

One would expect that, in a category of spectra with a symmetric monoidal structure modelling the smash product of the stable homotopy category, one could construct an analogous theory writing heuristically $A \wedge^{\mathbb{L}}_{A \wedge_{\mathbb{S}} A^{op}} A$, however when this is not enough. For example such a construction may not have the additional equivariant structure to define the appropriate generalisation of HC^{-} to spectra, (see the discussion in Chapter 4 of [DGM13], see also [DMP+19]).

While Böckstedt [Böc85] was the first to define an appropriate notion of the topological Hochschild homology and define the natural transformation from the K-theory spectrum of a ring to its topological Hochschild homology spectrum, it was in the work of Böckstedt-Hsiang-Madsen [BHM93], that the appropriate generalisation of $\mathrm{HC}^{-}(A)$ was successfully constructed, wherein the concept of cyclotomic spectra (Def. 2.17) was introduced. Cyclotomic spectra are a particular class of S^1-equivariant spectra which behave 'coherently' with respect to the geometric fixed points with respect to the finite subgroups of S^1. To every cyclotomic spectrum X, one can associate, functorially a topological cyclic homology $\mathrm{TC}(X)$ of that spectrum (Def. 2.18).

As it turns out, for a highly structured ring spectrum A, its topological Hochschild homology spectrum $\mathrm{THH}(A)$, when appropriately constructed will have enough equivariant structure to exhibit a cyclotomic structure. The topological cyclic homology of the topological Hochschild homology spectrum of A is called the topological cyclic homology of A, denoted $\mathrm{TC}(A)$. Then $\mathrm{TC}(A)$ is the correct integral spectrum refinement of negative cyclic homology $\mathrm{HC}^{-}(A)$ in case A was a discrete ring to begin with.

Theorem 1.1. *(Dundas–Goodwillie–McCarthy, [DGM13]) There is a trace map*

$$\mathrm{tr} : K(A) \to \mathrm{TC}(A)$$

[2]In the sense that the homotopy groups of $\mathrm{HH}(A)$ agree with the homology of the complex $\mathrm{HH}(A)$.

called the cyclotomic trace map, refining the Jones-Goodwillie trace, so that if $A \to \bar{A}$ is a map of connective, unital and highly structured ring spectra, with $\pi_0 A \to \pi_0 \bar{A}$ having nilpotent kernel, then the diagram

$$\begin{array}{ccc} K(A) & \xrightarrow{\mathrm{tr}} & \mathrm{TC}(A) \\ \downarrow & & \downarrow \\ K(\bar{A}) & \xrightarrow{\mathrm{tr}} & \mathrm{TC}(\bar{A}) \end{array}$$

is homotopy cartesian.

In order that the topological Hochschild homology of a highly structured ring spectrum has enough structure to exhibit a cyclotomic structure particular one has to work with a construction, called the Böcksted Construction (Def. 3.29), which is essentially a fibrant replacement of the bar construction for highly structured ring spectra. The obvious advantage of such a construction is that it has an explicit behaviour with respect to fixed points of the finite subgroups of S^1 (see Theorem 3.32).

At the time of the work of Böckstedt there was no category of spectra with a symmetric monoidal smash product. In particular to model rings upto coherent homotopy he was forced to build in the coherence conditions via a somewhat complicated machinery which he called Functors with Smash Products (FSP's). While from a modern viewpoint, these objects are very similar to the Symmetric spectra(for a precise comparison one should refer to [MMSS01]), the use of FSP's made the construction more complicated than it need be. For one thing, one had to put very complicated convergence conditions on the spectrum and for another one had to work at the level of spaces, one could only construct a space $\mathrm{THH}(L)$ for every FSP L, and to get the spectrum one would have to resort to an infinite loop space machine, for example like Segal's Γ-space delooping construction.

In this note, taking advantage of work done in [Shi00], [MMSS01], [BM15] and [NS18] and with hindsight of over three decades, we give a point set version of this construction by working in the category of orthogonal spectra.

This has the advantage of giving us a very concrete functorial construction which takes in an Eilenberg-Maclane orthogonal spectrum and outputs a cyclotomic orthogonal spectrum (see Def. 3.33).

In this expository note, we begin in Section 2 by reviewing the category of orthogonal spectra, particularly the equivariant story. Once we have set up the requisite fixed point functors we define cyclotomic orthogonal spectra.

We then start in Section 3 by recalling and defining some notions relevant for our study of cyclic objects in the category of orthogonal spectra. In particular in this section to study the Böcksted construction one has to define the associative operad $\mathrm{Ass}^{\otimes}$ at the level of categories so that functors out of it into a symmetric monoidal category $(\mathcal{C}, \otimes)$ can be identified with the associative algebra objects in that category (i.e. monoids with respect to $\otimes$). In particular this will be necessary to establish the functoriality of the Böckstedt construction. Also important to introduce are the Connes' category Λ and its variants, essentially an enhancement of the simplex category to include a cyclic automorphism, and their behaviour with respect to each other via simplicial subdivision, and with the associative operad $\mathrm{Ass}^{\otimes}$. Then we move onto define the Böckstedt construction and establish its good properties with respect to the fixed point functors of 2. Finally we apply the construction to define the THH of a ring spectrum in the category of orthogonal spectra.

We the end in Section 4 by giving an overview of the calculations of $\mathrm{TC}(\mathbb{F}_p)$ as done in the paper of Hesselholt-Madsen [HM97].

We would like to end this section with some words of advice on further reading. In [NS18] Nikolaus and Scholze have greatly simplified the construction of $\mathrm{TC}(A)$ for any associative ring spectra in the setting of Lurie and Joyal's quasi-categories (cf. [Lur17] and [Lur09]). This construction can be thought of as a refinement of the construction presented in this paper in the sense that the Dwyer-Kan localization of the category of orthogonal spectra and the constructions of cyclotomic spectra agree on bounded below objects (cf. Chapter III.6 in *loc.cit.*) Their construction offers a great simplification of the construction of TC and offers a streamline framework for carrying out hitherto complicated computations. We refer the reader to Hesselholt and Nikolaus survey [HN19] for further discussions on this point. Further reading in this direction is the Oberwolfach report [Obe18].

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2 Cyclotomic Spectra

In this section we will define Cyclotomic Spectra. For various reasons of technical ease as highlighted in the introduction we will be working in the category of Orthogonal spectra. Briefly, this point set construction of spectra allows us to be very explicit in our description of thee fixed points. Moreover one gets spectra level constructions and does not have to worry about various delooping constructions such as in [BHM93]. When one constructs Böckstedt's machinery (Section 3.29), one can functorially assign an orthogonal spectrum to finite sequence of spectra.

In this section we begin with a reminder on orthogonal spectra, and on equivariant orthogonal spectra. For both we follow Schwede's lectures on equivariant stable homotopy [Sch16]. We then define cyclotomic spectra following [NS18] and [BM15] and define cyclic homologyfollowing [HM97].

2.1 Reminders on Orthogonal Spectra

Definition 2.1. 1. An orthogonal spectrum X is the collection of

   (a) A pointed space X_n for all $n \geq 0$,
   (b) a continuous action of the orthogonal group $O(n)$ on X_n, preserving the basee point, and
   (c) base-point preserving maps $\sigma_n : X_n \wedge S^1 \to X_{n+1}$, subject to the condition that for all $n, m \geq 0$ the iterated structure maps

$$\sigma_{n+m-1} \circ (\sigma_{n+m-2}) \circ \ldots \circ (\sigma_n \wedge S^{m-1}) : X_n \wedge S^m \to X_{n+m},$$

   are $O(m) \times O(n)$ equivariant.[3]

2. Let X be an orthogonal spectrum and $i \in \mathbb{Z}$. The i-th stable homotopy group of X is

$$\pi_i(X) = \varinjlim_{\sigma} \pi_{i+n}(X_n).$$

3. A morphism $f : X \to Y$ of orthogonal spectra consists of $O(n)$-equivariant based maps $f_n : X_n \to Y_n$ for $n \geq 0$, which are compatible with the structure maps in the sense that $f_{n+1} \circ \sigma_n = \sigma_n \circ (f_n \wedge S^1)$ for all $n \geq 0$.
4. A map of orthogonal spectra $f : X \to Y$ is a stable equivalence if for all $i \in \mathbb{Z}$, the induced map $\pi_i f : \pi_i X \to \pi_i Y$ is an isomorphism.

From Def. 2.1 it is evident that orthogonal spectra naturally form a category which we denote Sp^O. In fact this category has a smash product, which we denote $\bigwedge_{\mathbb{S}}$ (cf. [Sch16, Section. 1] for details of the construction). This smash product gives Sp^O a symmetric monoidal structure and models the smash product of the stable homotopy category (see [MMSS01] for the relationship with other model categories of spectra).

In particular then, monoids with respect to this smash product are spectra which have an associative multiplication upto coherent homotopy, not just homotopy, analgous to the A_∞-ring spectra of [EKMM97]. In the same way, the commutative monoids with respect to this smash product are the spectra which have an associative and commutative multiplication upto coherent homotopy, analgous to the E_∞-ring spectra in *loc.cit*.

Example 2.2. We give two examples pertinent to our needs.

1. The sphere spectrum $\mathbb{S}$. In this case S^n arises as a one point compactification of $\mathbb{R}^n$ pointed at ∞. This has a natural action of $O(n)$ acting by isometries. The natural transition map $\sigma : S^n \wedge S^1 \to S^{n+1}$

[3] Let us note that the map $\sigma_n \wedge S^{m-1}$ is defined by first identifying S^m with $S^1 \times S^{m-1}$.

is an isomorphism. The homotopy groups are of course the stable homotopy groups of the spheres. The sphere spectrum is the initial commutative monoid in the category Sp^O (this explains our notation for the smash product).

2. The Eilenberg-MacLane spectrum associated to an abelian group A, is the spectrum $H(A)$ which is defined in degree n as the reduced A linearlisation of S^n denoted $A[S^n]$. More precisely HA_n is the tensor product of A with the free abelian group on S^n modulo the relation that coefficients of the base-point are 0. This space is topologized by the quotient topology induced by the canonical map $\coprod_{k\geq 0} A^k \times (S^n)^k \to A[S^n]$ which sends $(a_1, \dots, a_k, s_1, \dots, s_k) \mapsto (\sum_i a_i s_i)$. It is not too hard to show that $A[S^n]$ is an Eilenberg-MacLane space. The action of $O(n)$ is induced via the action of $O(n)$ on S^n and the canonical map $\sigma : S^1 \wedge HA_n \to HA_{n+1}$ is the adjoint of the equivalence $\tilde{\sigma} : HA_n \to \Omega HA_{n+1}$. It is actually not too hard (once one examines the construction of the smash product $\bigwedge_{\mathbb{S}}$) to see that if A is an ordinary associative (resp. commutative) ring, then the associated Eilenberg-Maclane spectrum HA is an associatve (resp. commutative) ring spectrum.

2.2 Equivariant Orthogonal Spectra

In this section we fix a finite group G.

Let BG be the associated groupoid for G. The notation is so chosen because the nerve of this category, $\mathrm{N}_\bullet(BG)$ is a simplicial set whose realization is the classifying space of G. For example if we set $G = C_p$, for some prime p, then $|\mathrm{N}_\bullet BC_p| = BC_p$.

For any category $\mathcal{C}$ the functor category $\mathrm{Fun}(BG, \mathcal{C})$ can be identified with the representations of G in that category. Given any object $c \in \mathcal{C}$, one has a natural homomorphism $G \to \mathrm{Aut}_{\mathcal{C}}(c)$. Of course this doesn't for This motivates our definition of the category of equivariant orthogonal spectra.

Definition 2.3. Let G be a finite group. The category of orthogonal G-spectra is the functor category $\mathrm{Fun}(BG, \mathrm{Sp}^O)$.

This definition is different from the usual definition of equivariant spectra given for example in [MM02]. We refer the reader to [Sch16, Rmk. 2.7] for a discussion of why these constructions are equal.

We will now describe a notion of geometric fixed points of a spectrum. For a finite dimensional G-representation V with $n = \dim_{\mathbb{R}}(V)$ and an orthogonal G spectrum X we define

$$X(V) = \mathcal{L}(\mathbb{R}^n, V)_+ \wedge_{O(n)} X_n \tag{2.1}$$

where $\mathcal{L}(\mathbb{R}^n, V)$ is the space of all linear isometries between $\mathbb{R}^n \to V$. We have the $\mathcal{L}(\mathbb{R}^n, V)$ is a torsor under the $O(n)$ action by precomposition. Moreover G acts on $X(V)$ under the diagonal action. One can now define a spectrum structure on $X(V)$ as V ranges through all finite dimensional representations of G. The way to do this is to define a map

$$\sigma_{V,W} : X(V) \wedge S^W \to X(V \oplus W), \tag{2.2}$$

see [Sch16, Eq. 2.4]. We note that there is an isomorphism $X_n \to X(\mathbb{R})$ mapping $x \mapsto [id, x]$. Thus these structure maps generalise the structure maps of equivariant orthogonal spectra. This point will be important as we go on.

Let ρ_G denote the regular representation $\mathbb{R}[G]$ of G, i.e. the vector space on an orthogonal basis labelled by G. We can apply the above construction to ρ_G itself in order to define the geometric fixed points of a spectrum.

Definition 2.4. [NS18, Def. II.2.3]

1. Let ρ_G denote the regular representation of G. Define an orthogonal spectrum $\Phi^G X$, calleed the geometric fixed points of X, whose n-th terem is given by

$$(\Phi^G X)_n = X(\mathbb{R} \otimes \rho_G)^G,$$

where thee fixed points on the RHS are the set theoretic fixed points.

2. Let $f : X \to Y$ be a map of orthogonal G-spectra. Then f is an equivalence if for all subgroups $H \subseteq G$, the map $\Phi^H X \to \Phi^H Y$ is a stable equivalence of orthogonal spectra.

We note that by [Sch16, Thm. 7.12] this agrees with the notion of π_* isomorphisms for spectra. Furthermore it is now obvious that the functor Φ^G sends equivariant spectra to non-equivariant spectra.

Remark 2.5. More generally $\Phi^G(X)(V) = X(V \otimes \rho_G)^G$ for a finite dimensional G representation V.

We also have

Proposition 2.6. *[Sch16, Prop. 7.14] The functor $\Phi^G : G\,\mathrm{Sp}^O \to \mathrm{Sp}^O$ is naturally lax symmetric monoidal. When either X or Y are cofibrant, then the natural map $\Phi^G X \wedge \Phi^G Y \to \Phi^G(X \wedge Y)$ is a stable equivalence.*

We would now like to define the notion of the naive fixed points.

Definition 2.7. For a G equivariant orthogonal spectrum X, the naive fixed points of X,denoted X^G is the spectrum which in degree n is just the fixed point of X_n under the G action i.e. $(X^G)_n = (X_n)^G$.

Here we will note an obvious compositional lemma for naive fixed points.

Lemma 2.8. *Let H be normal in G and X a G-equivariant orthogonal spectrum. Then*

$$(X^H)^{G/H} = X^G.$$

Proof. Follows from the space level statement. $\square$

Remark 2.9. Mostly, we will be working with orthgonal spectra having an action of S^1. In this case the definitions of the naive fixed points as in Def 2.7 goes through. We will in fact be working with fixed points of finite subgroups of S^1 and so the version of Lemma 2.15 that will be important to us will be $(X^{C_m})^{C_p} = X^{C_{mn}}$.

Remark 2.10. We now note that the the naive fixed points do not preserve weak equivalences between spectra. In fact they do so only for G-Ω-Spectra [Sch16, Def. 3.13], and so have to be right derived in the stable homotopy category.

2.3 Complete Universes and Geometric Fixed points.

In this section we let G be a compact Lie group.

Definition 2.11. [Sch18, Def. 3.1.7] Let G be a compact Lie group. A G-equivariant orthogonal spectrum is an orthogonal spectrum with a (level wise) continuous G-action.

For an equivalence with the definition given in, for example [MM02] see Remark 3.1.8. in *loc. cit.*

To proceed further we would like to look at geometric fixed points of finite (hence cyclic) subgroups C_p of S^1 where S^1 is the circle group. This will allow us to define a cyclotomic spectrum, which will be a S^1-equivariant (hence $RO(S^1)$ graded) orthogonal equivariant spectrum satsifying various compatibility conditions between the fixed points.

In order to do so we first note that the group S^1 has the exceptional property that the map sending $z \mapsto z^n$ defines an isomorphism $S^1/C_p \cong S^1$ of Lie groups. It is this construction which will be needed in particular to identify the C_p fixed points of an S^1 equivariant spectrum with an S^1 spectrum again using the heuristic principle that the C_p fixed points of an S^1-space X have a canonical action of S^1/C_p action and hence are once again S^1-spaces.

Given a finite subgroup H of a group G, and H-equivariant orthogonal spectrum X, we would like to define a residual G/H action on the fixed point spectra X^H and $\Phi^H X$.

In the case of X^H there is a canonical way to do so. However in the case of $\Phi^G X$ this is not so easy. Indeed there is no canonical way to extend the action of G on the regular representation ρ_H of H. We would like to rectify this by replacing ρ_H by a representation on which G acts. For this we will have to hinge on the notion of complete G-universes as defined in [LMSM86].

Definition 2.12. A complete G -universe is a representation $\mathcal{U}$ of G on a countable dimensional inner product $\mathbb{R}$ -vector space that is a direct sum of countably many copies of each irreducible representation of G.

Since we will be passing through G and associated normal subgroups, it is better to clarify the relationships the complete universes have with each other. If H is a subgroup of G, then a complete G-universe is canonically a complete H-universe. If H is normal in G, and $\mathcal{U}$ a complete G-universe then universe $\mathcal{U}^H$ of H fixed vectors is a G/H-universe.

Thus, if one has fixed a complete G-universe, one gets corresponding compatible complete G-universes for all subquotients of G, and in the following we will always fix a complete G-universe for the biggest group G around, and endow all other groups with their induced complete universes. In the following for a given universe $\mathcal{U}$ we write $V \subset \mathcal{U}$ if V is a finite dimensional G-sub-representation of U. These naturally form a poset.

The idea now is to replace ρ_H by a universal construction on which G-canonically acts.

Definition 2.13. Let G be a finite group with a complete G-universe $\mathcal{U}$, and let $H \subseteq G$ be a normal subgroup. The geometric fixed point functor

$$\Phi_{\mathcal{U}}^H : G\,\mathrm{Sp}^O \to (G/H)\mathrm{Sp}^O$$

is given by sending $X \in G\mathrm{Sp}^O$ to the orthogonal G/H -spectrum $\Phi_{\mathcal{U}}^H X$ whose n-th term is given by

$$\operatorname*{hocolim}_{V \in \mathcal{U}, V^H=0} X\left(\mathbb{R}^n \oplus V\right)^H$$

where $O(n)$ acts on $\mathbb{R}^n$, and the G-action factors over a G/H -action. The structure maps are the evident maps.

Here let us spell out the evident maps. For the poset $V \in \mathcal{U}$ with $V^H = 0$ we have in each level of the poset the morphism $\sigma : X(\mathbb{R}^n \oplus V) \wedge S^1 \to X(\mathbb{R} \oplus \mathbb{R}^n \oplus V)$ which exist because of equation 2.2. These pass to morphisms on the homotopy colimit.

First we would like a proposition which directly compares the above construction of Def. 2.13 with those of Def. 2.4.

We define

$$\widetilde{\Phi}_{\mathcal{U}}^G X := \operatorname*{hocolim}_{V \in \mathcal{U}, V^H=0} X\left(\mathbb{R}^n \otimes \rho_G \oplus V\right)^G. \tag{2.3}$$

Lemma 2.14. *Let G be finite. For every G -spectrum X there is a zig-zag of stable equivalences of orthogonal spectra $\Phi^G X \to \widetilde{\Phi}_{\mathcal{U}}^G X \leftarrow \Phi_{\mathcal{U}}^G X$ which is natural in X.*[17]

Proof. There is an obvious map $\Phi^G \to \widetilde{\Phi}_{\mathcal{U}}^G X$ mapping into the homotopy colimit. For the $\Phi_{\mathcal{U}}^G X \to \widetilde{\Phi}_{\mathcal{U}}^G X$ via the map sending $\mathbb{R} \to (\rho_G)^G$ mapping $1 \to \frac{1}{\sqrt{G}} \sum_{g \in G} g$. The stable equivalences are then checked by passing onto the homotopy groups, where they are well known. □

The essential use of Def. 2.13 is that these fixed point functors behave well with respect to composition.

Proposition 2.15. *Let G be a finite group with complete G-universe $\mathcal{U}$, and let $H \subseteq H' \subseteq G$ be normal subgroups. There is a natural transformation $\Phi_{\mathcal{UH}}^{H'/H} \circ \Phi_{\mathcal{U}}^H \to \Phi_{\mathcal{U}}^{H'}$ of functors $G\mathrm{Sp}^O \to (G/H')\,\mathrm{Sp}^O$. For each $X \in G\mathrm{Sp}^O$, the map*

$$\Phi_{\mathcal{U}^H}^{H'/H}\left(\Phi_{\mathcal{U}}^H(X)\right) \to \Phi_{\mathcal{U}}^{H'}(X)$$

is an equivalence of orthogonal (G/H') -spectra.

Proof. This follows from a simple computation with homotopy colimits. See [NS18, Prop. II 2.12] for details. □

2.4 S^1-Spectra and Cyclotomic Spectra

We want to now define S^1-equivariant orthogonal spectra and $\mathcal{F}$ equivalences, where $\mathcal{F}$ is the set of finite groups $C_p \subset S^1$ as n varies. Specialising Def. 2.11 we have,

Definition 2.16. The category $S^1\mathrm{Sp}^O$ of orthogonal S^1 -spectra is the category of orthogonal spectra with continuous S^1-action. Let $\mathcal{F}$ be the set of finite subgroups $C_n \subseteq S^1$.

A map $f : X \to Y$ of orthogonal S^1 -spectra is an $\mathcal{F}$ -equivalence if the induced map of orthogonal C_n -spectra is an equivalence for all finite subgroups $C_n \subseteq S^1$.

Our next objective is to introduce the category of cyclotomic spectra in the sense of [HM97, Sec. 2]. The main objective of this construction is that when we construct the topological Hochschild homology of a ring spectrum comes with a cyclotomic structure. This particular structure will allow us to define cyclic homology. It is now useful to pick a particular complete S^1 universe $\mathcal{U}$ as below.

$$\mathcal{U} = \bigoplus_{k \in \mathbb{Z}, i \geq 1} \mathbb{C}_{k,i}$$

where S^1 acts on $\mathbb{C}_{k,i}$ via the k -th power of the embedding $S^1 \hookrightarrow \mathbb{C}^\times$. For this universe, we have an identification

$$\mathcal{U}^{C_n} = \bigoplus_{k \in n\mathbb{Z}, i \geq 1} \mathbb{C}_{k,i}$$

which is a representation of S^1/C_n. Now there is a natural isomorphism

$$\mathcal{U}^{C_n} \cong \mathcal{U}$$

which sends the summand $\mathbb{C}_{k,i} \subseteq \mathcal{U}^{C_n}$ for $k \in n\mathbb{Z}$ and $i \geq 1$ identically to the summand $\mathbb{C}_{k/n,i} \subseteq \mathcal{U}$. This isomorphism is equivariant for the $S^1/C_n \cong S^1$ -action given by the n -th power. From Def.2.13 we get functors

$$\Phi_{\mathcal{U}}^{C_n} : S^1\mathrm{Sp}^O \to \left(S^1/C_n\right)\mathrm{Sp}^O \simeq S^1\mathrm{Sp}^O.$$

We now know that from Prop. 2.15 these functors have to satisfy the following coherent $\mathcal{F}$ equivalences.

$$\Phi_{\mathcal{U}}^{C_m} \circ \Phi_{\mathcal{U}}^{C_n} \to \Phi_{\mathcal{U}}^{C_{mn}}$$

Definition 2.17 (Cyclotomic Spectra)**.** An orthogonal cyclotomic spectrum is an object $X \in S^1\mathrm{Sp}^O$ together with $\mathcal{F}$-equivalences of orthogonal S^1-spectra $\Phi_n : \Phi_{\mathcal{U}}^{C_n} X \to X$ in $S^1\mathrm{Sp}^O \simeq \left(S^1/C_n\right)\mathrm{Sp}^O$ for all $n \geq 1$, such that for all $m, n \geq 1$, the diagram commutes.

$$\begin{tikzcd}
\Phi_{\mathcal{U}}^{C_n}(\Phi_{\mathcal{U}}^{C_m} X) \arrow[r, "\cong"] \arrow[d, "\Phi_{\mathcal{U}}^{C_n}(\Phi_m)"'] & \Phi_{\mathcal{U}}^{C_{mn}} X \arrow[d, "\Phi_{mn}"] \\
\Phi_{\mathcal{U}}^{C_n} X \arrow[r, "\phi_n"] & X
\end{tikzcd}$$

A map $f : X \to Y$ of orthogonal cyclotomic spectra is an equivalence if it is an $\mathcal{F}$ -equivalence of the underlying object in TSp^O. Let CycSp^O denote the category of orthogonal cyclotomic spectra.

2.5 Topological cyclic homology

In this section we will describe the Topological cyclic homology associated to a cyclotomic spectrum (Def. 2.17) following [HM97, Sec. 4.1]. We will first fix the complete S^1 universe $\mathcal{U}$ as in Section 2.4. We will fix an orthogonal cyclotomic spectrum X.

We first consider the category $\mathbb{I}$ where objects are the natural numbers, $\mathrm{Ob}\,\mathbb{I} = \{1, 2, 3, \ldots\}$, and with two morphisms $R_r, F_r : n \to m$, whenever $n = rm$ subject to the relations

$$\begin{aligned}
R_1 &= F_1 = \mathrm{id}_n \\
R_r R_s &= R_{rs}, \; F_r F_s = F_{rs} \\
R_r F_s &= F_s R_r
\end{aligned}$$

We will now show how a cyclotomic orthogonal spectrum X defines a functor from $\mathbb{I}$ to the category of non-equivariant spectrum. For $n \in \mathrm{Ob}(\mathbb{I})$ we associate the spectrum X^{C_p} where $(-)^{C_p}$ is the functor of Def. 2.7.

Fix an $n, m, r \in \mathrm{Ob}(\mathbb{I})$ with $n = rm$. We now define a Frobenius morphism F_r morphism. This is just the inclusion of fixed point

$$F_r : X^{C_p} \to X^{C_m}.$$

We will now define the Restriction map R_r as follows. By Remark 2.9 we can write X^{C_p} as $(X^{C_r})^{C_m}$. Since the trivial representation embeds in the regular representation of a finite group H there is always a natural transformation $(-)^H \to \Phi^H(-)$ (here $\Phi^H(-)$ is the fixed point functor of Def. 2.4) and moreover by Lemma 2.14, passes to a natural transformation $(-)^H \to \Phi^H_{\mathcal{U}}(-)$. Specialising to the case of $H = C_r$ we get a morphism

$$X^{C_r} \to \Phi^{C_r}_{\mathcal{U}}(X).$$

However by the definition of a cyclotomic spectrum there is an equivalence

$$\Phi_r : \Phi^{C_r}_{\mathcal{U}}(X) \simeq X.$$

Composing with the previous map and taking C_m fixed points (note that there is an action of C_m on C_r fixed points, since it is still an S_1-equivariant spectrum) we define R_r as the composition

$$R_r : X^{C_p} = (X^{C_r})^{C_m} \to (\Phi^{C_r}_{\mathcal{U}}(X))^{C_m} \simeq (X)^{C_m}.$$

One can check that these maps satisfy the commutation results.

We will now define topological cyclic homology TC and various variants, TF and TR. For calculation purposes it will be useful to describe a 'local' version as well. For this , if p is a prime we consider the full subcategory $\mathbb{I}_p$ with $\mathrm{Ob}(\mathbb{I}_p) = \{1, p, p^2, \dots, \}$. Indexed by $\mathbb{I}_p$ one gets two towers as follows

$$\dots X^{C_p^n} \xrightarrow{F_p} X^{C_p^{n-1}} \xrightarrow{F_p} X^{C_p^{n-2}} \xrightarrow{F_p} \dots,$$

and

$$\dots X^{C_p^n} \xrightarrow{R_p} X^{C_p^{n-1}} \xrightarrow{R_p} X^{C_p^{n-2}} \xrightarrow{R_p} \dots.$$

Definition 2.18. Let X be an orthogonal cyclotomic spectrum and p a prime.. We define the topological cyclic homology of X at p as

$$\mathrm{TC}(X;p) = \underset{\overleftarrow{\mathbb{I}_p}}{\mathrm{holim}}\ X^{C_{p^s}},$$

where the homotopy inverse limit is taken over the maps F_p and R_p. The topological cyclic homology of X is defined as

$$\mathrm{TC}(X) = \underset{\overleftarrow{\mathbb{I}}}{\mathrm{holim}}\ X^{C_n}.$$

The inclusions $\{1\} \subset \mathbb{I}_p \subset \mathbb{I}$ induce maps

$$\mathrm{TC}(X, p) \to \mathrm{TC}(X) \to X.$$

The following theorem of Goodwillie (cf. [HM97, Sec. 4.5]) states that $\mathrm{TC}(X)$ is not a stronger functor than the collection of $\mathrm{TC}(X;p)$ for all $p's$.

Theorem 2.19 (Goodwillie)**.** *The projections* $\mathrm{TC}(X) \to \mathrm{TC}(X;p)$ *induce an equivalence of* $\mathrm{TC}(X)$ *with the fiber product of the* $\mathrm{TC}(X;p)$*'s over X. Moreover, the functors agree after p-adic completion,* $\mathrm{TC}(X)^\wedge_p \simeq \mathrm{TC}(X;p)^\wedge_p$.

Proof. See [HM97, Sec. 4.5]. ☐

Remark 2.20. This result will be particularly useful in the case of cyclotomic spectra so that $\mathrm{TC}(X)$ and $\mathrm{TC}(X;p)$ is already p-complete, for example $\mathrm{TC}(\mathbb{F}_p)$ with $\mathbb{F}_p$ the prime field of char $p > 0$. as in Def. 3.38.

Now we define some other variants of TC which are sometimes useful in calculations.

Definition 2.21. Let p be a prime and R_p and F_p as above and X the orthogonal cyclotomic spectrum.

1. $\mathrm{TR}(X;p)$ is defined as $\underset{\overleftarrow{R_p}}{\mathrm{holim}}\ X^{C_{p^s}}$, and $\mathrm{TR}(X)$ as $\underset{\overleftarrow{R}}{\mathrm{holim}}\ X^{C_n}$.
2. $\mathrm{TF}(X;p)$ is defined as $\underset{\overleftarrow{F_p}}{\mathrm{holim}}\ X^{C_{p^s}}$, and $\mathrm{TF}(X)$ as $\underset{\overleftarrow{F}}{\mathrm{holim}}\ X^{C_n}$.

Now note that F_p (resp. R_p) still acts on $\mathrm{TR}(X;p)$ (resp.$\mathrm{TR}(X;p)$). Then consider the free monoid $\langle F_p\rangle$ (resp. $\langle R_p\rangle$). Then one can define

$$\mathrm{TC}(X;p) = \mathrm{TR}(X;p)^{h\langle F_p\rangle} = \mathrm{TF}(X;p)^{h\langle R_p\rangle}.$$

Here $h\langle F_p\rangle$ (resp. $h\langle R_p\rangle$) denote the homotopy fixed points of the monoid action of F_p(resp. R_p). This is how cyclic homology was originally introduced in [BHM93]. There is of course a straightforward global version for $\mathrm{TC}(X)$ as well.

$$\mathrm{TC}(X) = \mathrm{TR}(X)^{h\langle F\rangle} = \mathrm{TF}(X)^{h\langle R\rangle},$$

where hF (resp. hR) denotes the homotopy fixed set of the multiplicative monoid of natural numbers acting of $\mathrm{TR}(X)$ (resp. $\mathrm{TF}(X)$) through the maps $F_s, s \geq 1$ (resp. $R_s, s \geq 1$).

Remark 2.22. We remark now that this definition of TC is has recently been refined. Using work on the co-representability of TC and TR in [BM15], Nikolaus and Scholze ([NS18, Def. II I.8]) define TC as a mapping spectrum in certain 'enriched' categories of spectra. This has the advantage that $\mathrm{TC}(X)$ can be defined as the homotopy fiber of certain easily computed invariants (Prop. II 1.9 in *loc. cit.*). This definition agrees with our definition (Def. 2.18) in case of bounded below spectra. In particular then one can calculate the TC of certain rings by conveniently foregoing all the equivariant stable homotopy theory involved.

3 Topological Hochschild Homology

In this section we define Topological Hochschild homology using the Böcksted construction and prove it is cyclotomic. First we will need to develop some machinery in order to handle all the information needed to define THH as a cyclotomic spectrum. In particular we need to describe Associative algebra objects in a category as functors from the associative operad $\mathrm{Ass}^{\otimes}$ in Construction 3.1 to the category. This is basically a cateegorification of Segal's Γ spaces in [Seg74]. This will have an advantage on making our construction obviously functorial and once we define the Böcksted construction, one will immediately get a natural map betweeen fixed points.

We also need to recall Connes category in Construction 3.12 and discuss various variants 3.11 and various functors among them 3.14. We also need the notion of simplicial subdivision 3.22.

After the preliminaries we define the Böckstedt construction as a functorial construction which takes a finite number of orthogonal spectra functorially to an orthogonal spectrum. To define this we need to introduce the construction 3.5.

After this we define THH as a special case of the Böckstedt construction and prove that it is cyclotomic.

3.1 Preliminaries on operads and combinatorial categories

Most of this has been worked out in Appendix B of [NS18]. We note that this is nothing new, and was known as far back as the work of Segal in [Seg74]; one can also consult the work of May and Thomason [MT78] on the equivalence of all these operadic approaches to delooping. One should understand this section as a necessary evil to make precise all the functorialities present in the Böckstedt construction. Moreover this section is also necessary to construct the correct homotopy type for the topological Hochschild homology of an orthogonal ring spectrum, in particular to show that the spectrum is indeed cyclotomic.

To begin, we denote by Fin_* the category of finite pointed sets and morphisms of finite pointed sets. These are objects $\langle n\rangle := \{0, 1, \ldots, n\}$ pointed at 0.

Definition 3.1 (Associative Operad)**.** We consider the operad $\mathrm{Ass}^{\otimes}$. It is a category consisting of finite pointed sets. Morphisms between these sets are morphisms of finite pointed sets along with a linear ordering on the preimages. Compositions for two maps $g : S \to T$ and $h : T \to W$ has a linear ordering on $a, b \in (h \circ g)^{-1}(i)$ by setting $a <_{h \circ g} b$ if $g(a) = g(b)$ and $a <_h b$ or if $g(a) <_h g(b)$. Since every object is in fact upto an isomorphism of the form $\langle n \rangle \in \mathrm{Fin}_*$ we will sometimes write objects in $\mathrm{Ass}^{\otimes}$ as $\langle n \rangle_{\mathrm{Ass}}$.

Thus there is a natural functor $F : \mathrm{Ass}^{\otimes} \to \mathrm{Fin}_*$ forgetting the linear order on the maps.

Definition 3.2. A morphism $f : \langle n \rangle \to \langle m \rangle$ in Fin_* is called inert if for every $0 \neq i \in \langle m \rangle$ the preimage $f^{-1}(i)$ contains exactly one element. A morphism $f : \langle m \rangle \to \langle n \rangle$ in Fin_* is active if $f^{-1}\{0\} = \{0\}$.

It is easy to see that every morphism f in Fin_* admits a unique factorization as $f = f' \circ f$" where f" is inert and f' is active.

Construction 3.3. Let Fin denote the category of finite sets with no distinguished points. There is a natural inclusion functor $\mathrm{Fin} \to \mathrm{Fin}_*$ which adds a distinguished base point. This functor defines an embedding of Fin into Fin_* as a subcategory consisting all objects and active maps. If $\langle n \rangle$ is an object of Fin_* we write view $\langle n \rangle^{\circ} = \langle n \rangle \setminus \{0\}$. We have a 2-fibre product of categories $\mathrm{Ass}^{\otimes}_{act} := \mathrm{Ass}^{\otimes} \times_{\mathrm{Fin}_*} \mathrm{Fin}$. Here objects are triples $(\langle n \rangle_{\mathrm{Ass}}, \langle n \rangle, f)$ where f is an isomorphism between S and $\langle n \rangle$ obtained upon applying the forgetful (resp. inclusion) functors into Fin_*. In particular this shows that $\mathrm{Ass}^{\otimes}_{act}$ can be identified with the subcategory of $\mathrm{Ass}^{\otimes}$ consisting of active morphisms, hence the name.

Construction 3.4. Let $\mathbf{\Delta}$ denote the simplex category i.e. finite ordered sets and non-decreasing maps. There is a natural functor $\mathbf{\Delta}^{op} \to \mathrm{Fin}_*$ which sends every ordered set to the set of cuts on that set. For a finite ordered set S the set of cuts on S, denoted $\mathrm{Cut}(S)$ is the number of disjoint decompositions $S_1 \sqcup S_2$ on S so that $\sup(S_1) < \inf(S_2)$. We identify the cuts $S \sqcup \emptyset$ and $\emptyset \sqcup S$. Then for $[n] = \{0 < 1 < \ldots < n\} \in \mathbf{\Delta}$, we have that the sets of cuts is isomorphic to $\langle n \rangle$ and pointed at the trivial cut $\emptyset \sqcup S$. The morphisms obviously get mapped to taking preimages.

There is a functor from $\mathbf{\Delta}^{op} \to \mathrm{Ass}^{\otimes}$ over Fin_*. Note that for $S \in \mathbf{\Delta}$ we get sent to $\mathrm{Cut}(S)$ in Fin_* where a morphism $f : S \to T$ gets mapped to $f^{-1} : \mathrm{Cut}(T) \to \mathrm{Cut}(S)$. This means we have to describe a linear ordering on cuts on T pulling back to cuts on S. In order to do this we have that for a cut $S_1 \sqcup S_2$ the pullbacks define a partition of T into two subsets. For two partitions $T_1 \sqcup T_2 = T$ is less are equal to a partition $T_1' \sqcup T_2' = T$ if $T_1 \subseteq T_1'$ and consequently $T_2 \supseteq T_2'$. This gives the required functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathrm{Ass}^{\otimes}$.

We now define a new notion of a symmetric monoidal category. This definition is only useful for us to define associative algebra objects as in [NS18].

Construction 3.5. [Lur17, Def. 2.0.0.1] Let$(\mathcal{C}, \otimes)$ be a symmetric monoidal category. We define a new category $\mathcal{C}^{\otimes}$ as follows:

1. An object of $\mathcal{C}^{\otimes}$ is a finite (possibly empty) sequence of objects of $\mathcal{C}$, which we will denote by $[C_1, \ldots, C_n]$
2. A morphism from $[C_1, \ldots, C_n]$ to $[C_1', \ldots, C_m']$ in $\mathcal{C}^{\otimes}$ consists of a subset $S \subseteq \{1, \ldots, n\}$, a map of finite sets $\alpha : S \to \{1, \ldots, m\}$, and a collection of morphisms $\left\{f_j : \bigotimes_{\alpha(i)=j} C_i \to C_j'\right\}_{1 \leq j \leq m}$ in the category $\mathcal{C}$. (Here the tensor product $\bigotimes_{\alpha(i)=j} C_i$ is well-defined up to canonical isomorphism, since $\mathcal{C}$ is a symmetric monoidal category.)
3. Suppose we are given morphisms $f : [C_1, \ldots, C_n] \to [C_1', \ldots, C_m']$ and $g : [C_1', \ldots, C_m'] \to [C_1'', \ldots, C_l'']$ in $\mathcal{C}^{\otimes}$, determining subsets $S \subseteq \{1, \ldots, n\}$ and $T \subseteq \{1, \ldots, m\}$ together with maps $\alpha : S \to \{1, \ldots, m\}$ and $\beta : T \to \{1, \ldots, l\}$. The composition $g \circ f$ is given by the subset $U = \alpha^{-1} T \subseteq \{1, \ldots, n\}$, the composite map $\beta \circ \alpha : U \to \{1, \ldots, l\}$, and the maps
$$\bigotimes_{(\beta \circ \alpha)(i)=k} C_i \cong \bigotimes_{\beta(j)=k} \bigotimes_{\alpha(i)=j} C_i \to \bigotimes_{\beta(j)=k} C_j' \to C_k''$$

Construction 3.6. If $\alpha : \langle n \rangle \to \langle m \rangle$ is a morphism in Fin_* it is convenient to think of α as a partially defined map from $\langle n \rangle^{\circ}$ to $\langle m \rangle^{\circ}$ in Fin namely, α is given by specifying a subset $S = \alpha^{-1}\langle m \rangle \subseteq \langle n \rangle^{\circ}$ together with a map $S \to \langle m \rangle^{\circ}$

Clearly there is a functor $\mathcal{C}^{\otimes} \to \mathrm{Fin}_*$ which sends every finite sequence of objects of $\mathcal{C}^{\otimes}$ to the corresponding object of Fin_* . We can uniquely expand every partially defined morphism by sending everything else to 0. Thus this construction gives us a functor $p : \mathcal{C}^{\otimes} \to \mathrm{Fin}_*$. This functor satisfies two important properties

1. This functor is in fact an op-fibration of categories in the sense of [GR71] (see also [Vis07] for a modern exposition). This in particular means that if $f : \langle m\rangle \to \langle n\rangle$ is a function in Fin_*, then one gets a functor $f_* : \mathcal{C}_{\langle m\rangle} \to \mathcal{C}_{\langle m\rangle}$. Here $\mathcal{C}^{\otimes}_{\langle n\rangle}$ is the fibre above $\langle n\rangle$ of the functor $p : \mathcal{C}^{\otimes} \to \mathrm{Fin}_*$.
2. The functor $p : \mathcal{C}^{\otimes} \to \mathrm{Fin}_*$ also satisfies the condition that $\mathcal{C}^{\otimes}_{\langle 1\rangle}$ is equivalent to $\mathcal{C}$ and moreover $\mathcal{C}^{\otimes}_{\langle n\rangle}$ is equivalent to the n-fold product of copies of $\mathcal{C}$. The latter equivalence is induced by the set of morphisms $\left\{\rho^i : \langle n\rangle \to \langle 1\rangle\right\}_{1\leq i\leq n}$ for i which for each i are defined by sending $j \mapsto 1$ if $j = i$ and to 0 otherwise. This is an analogue of the 'Segal' condition in [Seg74].

Definition 3.7 (Associative Algebra Objects)**.** [NS18, Def. III.2.1] Let $(\mathcal{C}, \otimes)$ be a symmetric monoidal category. Let $\mathcal{C}^{\otimes} \to \mathrm{Fin}_*$ be the associated op-fibration as in Construction 3.5. The category $\mathrm{Ass}_{\mathrm{Alg}}(\mathcal{C})$ of associative algebras in the symmetric monoidal category $(\mathcal{C}, \otimes)$ is given by the category of functors $\mathrm{Ass}^{\otimes} \to \mathcal{C}^{\otimes}$ over Fin_* which carry inert maps to inert maps.

The final condition essentially says that for all $n \geq 1$ the image of the unique object $\langle n\rangle_{\mathrm{Ass}}$ in $\mathrm{Ass}^{\otimes}$ mapping to $\langle n\rangle \in \mathrm{Fin}_*$ is given by $(A, \dots, A) \in \mathrm{Sp}^n$, where $A \in \mathrm{Sp}$ is the image of $\langle 1\rangle_{\mathrm{Ass}}$; this condition is spelled out more precisely in the next proposition.

Proposition 3.8. *Let $\mathcal{C}^{\otimes}$ be as in Def. 3.7, and let $\mathbf{\Delta}^{op} \to \mathrm{Ass}^{\otimes}$ be the functor in Construction 3.4. Then restriction of the functor category in Def. 3.7 along this functor defines an equivalence between $\mathrm{Ass}_{\mathrm{Alg}}(\mathcal{C})$ and the category of functors*

$$\begin{tikzcd} \mathbf{\Delta}^{op} \arrow[r, "A^{\otimes}"] \arrow[dr] & \mathcal{C}^{\otimes} \arrow[d] \\ & \mathrm{Fin}_* \end{tikzcd}$$

making the diagram commute, and satisfying the 'Segal'-condition of Construction 3.6.

Proof. See [NS18, Prop. III 2.2]. ◻

Construction 3.9. In the spirit of Construction 3.3, we can construct the category $\mathcal{C}^{\otimes}_{act} := \mathcal{C}^{\otimes} \times_{\mathrm{Fin}_*} \mathrm{Fin}$ by taking fibre products along the functor $p : \mathcal{C}^{\otimes} \to \mathrm{Fin}_*$ constructed in Construction 3.6. This is the subcategory of $\mathcal{C}^{\otimes}$ where morphism from $[C_1, \dots, C_n]$ to $[C'_1, \dots, C'_m]$ in $\mathcal{C}^{\otimes}$ consists of a map of finite sets $\alpha : \{1, \dots, n\} \to \{1, \dots, m\}$, and a collection of morphisms $\left\{f_j : \bigotimes_{\alpha(i)=j} C_i \to\ C'_j\right\}_{1\leq j\leq m}$ in the category $\mathcal{C}$ (constrast with (2) in Construction 3.5).

Construction 3.10. Given a group G, one constructs a groupoid BG, the nerve of which is homotopy equivalent to the classifying space of G. Given an arbitrary (small) category $\mathcal{C}$, an action of G on $\mathcal{C}$ is a pseudofunctor $BG \to \mathrm{Cat}$ with image $\mathcal{C}$ (see [Vis07] for the definition of a pseudofunctor). We abuse notation and write this as an action of BG on $\mathcal{C}$.

Given two categories $\mathcal{C}$ and $\mathcal{D}$ a BG equivariant functor $F : \mathcal{C} \to \mathcal{D}$ is a functor so that for all $g \in \mathrm{Hom}_{BG}(*, *)$ the diagram of functors

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] \arrow[d, "g_*"'] & \mathcal{D} \arrow[d, "g_*"] \\ \mathcal{C} \arrow[r, "F"] & \mathcal{D} \end{tikzcd},$$

2-commutes i.e. $F \circ g_* \cong g_* \circ F$. Another notion which will be specifically useful for our purposes is the following. For any category $\mathcal{C}$, the subcategory of G-equivariant objects is the functor category $\mathrm{Fun}(BG, \mathcal{C})$. There is a canonical action of BG on $\mathrm{Fun}(BG, \mathcal{C})$ induced by the action of BG on itself, which is given by the action on the set of natural transformations, i.e. the morphism sets of $\mathcal{C}$.

Definition 3.11 (Paracyclic Category)**.** Let PoSet be the category of posets with non-decreasing maps. Let $\mathbb{Z}$Poset be the subcategory of $\mathbb{Z}$ equivariant objects in Poset. Then the paracyclic category, Λ_∞ is the full subcategory of $\mathbb{Z}$Poset isomorphic to $\frac{1}{n}\mathbb{Z}$ with its canonical ordering and $\mathbb{Z}$ action given by addition. We denote by $[n]_{\Lambda_\infty}$ an object $\frac{1}{n}\mathbb{Z}$ of Λ_∞.

Construction 3.12 (Connes' Category)**.** There is an action of $B\mathbb{Z}$ on Λ_∞, given by the restriction of the action on $\mathbb{Z}\mathrm{PoSet} = \mathrm{Fun}(B\mathbb{Z}, \mathrm{PoSet})$. Parsing the definitions this means that there is an action of $\mathbb{Z}$ on the morphism sets of Λ_∞ where in a generator of σ of $\mathbb{Z}$ acts by sending $f \mapsto f + 1$. We denote new quotient categories $\Lambda_p = \Lambda_\infty/B(p\mathbb{Z})$ for $p \geq 1$. Briefly in these categories we have the same objects but the morphism sets come divided out by the action of σ^p for σ the generator of $\mathbb{Z}$. In particular we have $\Lambda := \Lambda_1 = \Lambda_\infty/B\mathbb{Z}$ which is the Connes' Category.

For a given category Λ_p with $1 \leq p \leq \infty$ we write an object $\frac{1}{n}\mathbb{Z}$ as $[n]_{\Lambda_p}$.

Construction 3.13 (Self-Duality)**.** It is easy to see that in fact Λ_∞ is self dual. Indeed for any morphism $f : S \to T$ we may define f° via the formula

$$f^\circ(x) := \min\{\ y \mid y \geq f(x)\ \}.$$

Note that this formula is $\mathbb{Z}$ equivariant and non-decreasing. We can recover f from f° by the formula

$$f(y) = \max\{\ x \mid x \leq f(y)\}.$$

Thus it is clear that $\Lambda_\infty^{op} \simeq \Lambda_\infty$ and since this equivalence is $B\mathbb{Z}$ equivariant, we get a self-duality of Λ_p for all $1 \leq p \leq \infty$, in particular of Λ.

Construction 3.14 (Functors between combinatorial categories)**.** Note that morphisms in Λ are identified upto action of $\mathbb{Z}$. Thus there is a natural quotient functor $V : \Lambda \to \mathrm{Fin}$ which sends $[n]_\Lambda \to [n]_\Lambda/\mathbb{Z} = \{0, \frac{1}{n}, \dots, \frac{n-1}{n}\}$. It is easy to see that this construction is functorial for Λ since $f \sim f + k$ for all $k \in \mathbb{Z}$. This functor is in fact on the nose $V(-) = \mathrm{Hom}_\Lambda([1]_\Lambda, -)$.

This functor lifts to a functor $V(-) : \Lambda \to \mathrm{Ass}^{\otimes}_{act} = \mathrm{Ass}^{\otimes} \times_{\mathrm{Fin}_*} \mathrm{Fin}$. This is not hard to see, and can be done by showing a linear ordering on the pre-image of $V(f)^{-1}(i)$ for every morphism $[n]_\Lambda \to [m]_\Lambda$ and $i \in V([n])$. This is easy to see by choosing a lift of the function f and a lift of $i \in V([m]_\Lambda)$ and noting that that comes with a total order. It is easy to see that this lift is independent of the choice of the lift.

This implies that that V passes to a natural functor $V : \Lambda_{/[1]_\Lambda} \to \mathbf{\Delta}$ where each map $f : [n]_\Lambda \to [1]_\Lambda$ defines a natural ordering on $V([n]_\Lambda)$. We want to now claim that this construction is in fact an equivalence. But this is not hard. First note that there is a natural map $\mathbf{\Delta} \to \Lambda_\infty$ given by sending $[n-1] := \{0 < 1 \ldots < n-1\} \to \mathbb{Z} \times [n-1] = \frac{1}{n}\mathbb{Z}$ where we use lexicographic ordering and have a $\mathbb{Z}$ action on the first coordinate. Since $\mathbf{\Delta}$ has a final object $[0]$ this passes to a functor $(\Lambda_\infty)_{/[1]_{\Lambda_\infty}}$. One can pass to quotients $\Lambda_{/[1]_\Lambda}$ and check that this functor is an equivalence. By applying the forgetful functor we get a natural functor $\mathbf{\Delta} \to \Lambda$ sending $[n]$ to $[n+1]_\Lambda$, or equivalently a functor $j : \mathbf{\Delta}^{op} \to \Lambda^{op}$. Note that by the considerations above, this actually factors over a functor $j_\infty : \Delta^{op} \to \Lambda_\infty^{op}$.

Construction 3.15 (Geometric Realisations of simplicial spaces)**.** We define various notions of geometric realisation of simplical and cyclic spaces. To begin with we have to define the realisation functor

$$|-| : \mathrm{Fun}(\mathbf{\Delta}^{op}, \mathrm{Top}) \to \mathrm{Top}.$$

Once one puts a cosimplicial structuree on the topological simplicies $|\Delta^n|$ we ca define geometric realisation as the coend

$$|X_\bullet| = \int_{S \in \Delta} X_S \times \left|\Delta^S\right| = \left(\coprod_{n \in \mathbb{N}} X_n \times |\Delta^n|\right) / \sim .$$

To describe the cosimplicial structure consider the category Int of intervals. We say that a linearly ordered set is an interval if it has a minimal and a maximal element which are distinct. This is based on the canonical example of $[0, 1] \subset \mathbb{R}$. A morphism of intervals is a non-decreasing map that sends the minimal element to the minimal element and the maximal element to the maximal element.

We define a functor $-^\circ : \mathbf{\Delta} \to \mathrm{Int}$. For $S \in \mathbf{\Delta}$, this is defined as

$$S^\circ = \mathrm{Hom}_{\mathbf{\Delta}}(S, [1]) = \{S = S_0 \sqcup S_1 | s \leq t \text{ for } s \in S_0, t \in S_1\},$$

where S° has the natural ordering as a set of maps into a poset. It is easy to see that S° is infact an interval.

Now we set

$$|\Delta^S| = \mathrm{Hom}_{\mathrm{Int}}(S^\circ, [0, 1]).$$

Here $|\Delta^S|$ has the induced topology from $[0,1]$. If $S = [n]$, then $S^\circ = [n+1]$. The point of this construction is that the functoriality for non-decreasing maps $[m] \to [n]$ is obvious. One can check that this realisation preserves the description of $|\Delta^n|$ as the set

$$|\Delta^n| = \{(y_1, \ldots, y_n) \in [0,1]^n | y_1 \leq \ldots \leq y_n\}.$$

Construction 3.16 (Geometric realisation of paracyclic spaces)**.** We now give a description of the geometric realisation of a paracyclic space by giving a $B\mathbb{Z}$ equivariant functor

$$| - | : \mathrm{Fun}(\Lambda_\infty^{op}, \mathrm{Top}) \to \mathbb{R}\mathrm{Top},$$

where the right side is the subcategory of all topological spaces with a continuous action of $\mathbb{R}$ and equivariant maps.

First recall the self duality functor which sends $T \to T^\circ$. Note that T° is a poset. We also consider $\mathbb{R}$ with the natural ordering. Thus we get for $T \in \Lambda_\infty$, we define

$$\left|\Lambda_\infty^T\right| := \{f : T^\circ \to \mathbb{R} \mid f \text{ non-decreasing and } \mathbb{Z} \text{ -equivariant } \}.$$

We topologize it with the topology on the space of maps to $\mathbb{R}$ with its natural topology. There is an action of $\mathbb{R}$ given by post composition with translation.

Now for an arbitrary paracyclic topological space $X_\bullet$ we define the geometric realization by the coend

$$|X_\bullet| = \int_{T \in \Lambda_\infty} X_T \times \left|\Lambda_\infty^T\right| = \coprod_T \left(X_T \times \left|\Lambda_\infty^T\right|\right) / \sim$$

in $\mathbb{R}\mathrm{Top}$. This functor is $B\mathbb{Z}$-equivariant, since the functor $T \mapsto \left|\Lambda_\infty^T\right|$ is.

Construction 3.17 (Geometric Realisation of a cyclic space)**.** A cyclic space is an object of the category $\mathrm{Fun}(\Lambda^{op}, \mathrm{Top})$. By definition of Λ this is same as a $B\mathbb{Z}$ equivariant functor $\Lambda_\infty^{op} \to \mathrm{Top}$. This comes down to being a condition on the functor $X_\bullet : \Lambda_\infty^{op} \to \mathrm{Top}$ i.e. it must identify the image of functions f with that of $f + k$ for $k \in \mathbb{Z}$. Thus we can consider cyclic spaces as a full subcategory of paracyclic spaces.

Now since the realisation functor of Construction 3.16 is $B\mathbb{Z}$ equivariant, we see that the realisation of a cyclic space $|X_\bullet|$ has trivial $\mathbb{Z}$ action and so in particular passes to an action of $\mathbb{R}/\mathbb{Z} = S^1$. Thus the realisation of a cyclic space lands in $S^1\mathrm{Top}$, the category of topological spaces with a continuous S^1-action. Similarly for the variants Λ_p considered in Construction3.12 we have realisation functors $| - | : \mathrm{Fun}(\Lambda_p^{op}, \mathrm{Top}) \to \mathbb{R}/p\mathbb{Z}\mathrm{Top}$. Under the isomorphism $\mathbb{R}/p\mathbb{Z} \cong S^1$ we see that these spaces to have an S^1 action.

Example 3.18. Every object of Λ_∞ is of the form $S \times \mathbb{Z}$ with $S \in \mathbf{\Delta}$. More precisely we have that the image of the functor $j_\infty(S)$ (see Construction 3.14.). It can be show that there is a canonical isomorphism

$$|\Delta^S| \times \mathbb{R} \cong |\Lambda_\infty^{j_\infty(S)}|.$$

In fact for any $S \in \Lambda$ one can show canonically that

$$|\Delta^S| \times \mathbb{R}/\mathbb{Z} \cong |\Lambda^{j(S)}|$$

where the second realisation is that if a cyclic space.

Construction 3.19 (Realisation for a parayclic spectrum)**.** A paracyclic spectrum is an object of the category $\mathrm{Fun}(\Lambda_\infty^{op}, \mathrm{Sp}^O)$. Concretely for each $X_\bullet$ as a paracyclic spectrum would consist of (for a fixed spectrum degree) a paracyclic space. We can take it's realisation which gives us a topological space in each spectra degree. It is easy to see that by functoriality this is consistent with the structure maps and so the realisation functor $| - | : \mathrm{Fun}(\Lambda_\infty^{op}, \mathrm{Sp}^O) \to \mathrm{Sp}^O$ lands in the category $\mathbb{R}\mathrm{Sp}^O$ the category of spectra with continuous $\mathbb{R}$ action. Of course we can look at $B\mathbb{Z}$ equivariant functor and then we will land in $S^1\mathrm{Sp}^O$.

Remark 3.20. We refer the reader to example remark B.11 in [NS18] for connections between the classical realisations of a cyclic space shown for example in [BHM93] and in [Jon87].

Recall from Construction 3.14 that we have a functor $j_\infty : \mathbf{\Delta}^{op} \to \Lambda_\infty^{op}$. A paracyclic space is a functor $X_\bullet : \Lambda_\infty^{op} \to \mathbb{R}\mathrm{Top}$. By precomposing with j_∞ we get a simplicial set denoted $j_\infty^* X_\bullet$. We then have the following lemma.

Lemma 3.21. *For a paracyclic topological space X. the underlying space of the geometric realization $X_\bullet$ is naturally homeomorphic to the geometric realization of the underlying simplicial space $j_\infty^* X_\bullet$*

Proof. See [NS18, Lem. B.12]. □

Construction 3.22 (Simplicial subdivision)**.** Here we recall that a category $\mathcal{C}$ is called contractible if the nerve NC is contractible. Recall from Construction 3.12 that $\Lambda = \Lambda_\infty / B(p\mathbb{Z})$. This means that there is a natural functor $\Lambda_p \to \Lambda$. In fact we can describe Λ as the quotieent of Λ_p by the action of BC_p. Upon taking geometric realisations this results in a fibration $\Lambda_p \to \Lambda$ with fibre BC_p. We will now describe a functor which actually induces a homotopy equivalence on the geometric realisations via Quillen's theorem A (cf. [Qui73]), although we will not prove it.

It is constructed as follows. There is an endofunctor $\mathrm{sd}_p : \Lambda_\infty \to \Lambda_\infty$ sending $[n]_{\Lambda_\infty}$ to $[pn]_{\Lambda_\infty}$, and a map $f : \frac{1}{n}\mathbb{Z} \to \frac{1}{m}\mathbb{Z}$ to the map $\frac{1}{p} f(p \cdot -) : \frac{1}{pn}\mathbb{Z} \to \frac{1}{pm}\mathbb{Z}$. More generally for an arbitrary object $T \in \Lambda_\infty$ we introduce a new object $\mathrm{sd}_p(T) = \frac{1}{p}T$ with the same underlying order set as T and the action of $n \in \mathbb{Z}$ given by multiplication with pn

By $B\mathbb{Z}$ equivariance it will pass to a functor between the categories $\Lambda_p \to \Lambda_p$ for all $1 \leq p \leq \infty$ and composing with the canonical map $\Lambda_p \to \Lambda$ we get our equivalence. In fact observe that the functor sends $[n]_{\Lambda_p} \mapsto [pn]_\Lambda$.

Note that $\mathrm{sd}_p : \Lambda_p \to \Lambda$ defines a functor $\mathrm{sd}_p : \Lambda_p^{op} \to \Lambda^{op}$ in the obvious fashion (there is something to check, see [NS18, Prop. B.17]). Composing with a cyclic space $X_\bullet$ we get a subdivided cyclic space $\mathrm{sd}_p(X_\bullet)$. We have the following crucial result.

Lemma 3.23. *Let $X_\bullet$ and $\mathrm{sd}_p X_\bullet$ be as above. Then there is a natural S^1 equivariant homeomorphism*

$$|X_\bullet| \cong |\mathrm{sd}_p X_\bullet|$$

Remark 3.24. Both Lemma 3.23 and Lemma 3.21 work in the case of Construction 3.19.

Construction 3.25 (Compatiblity of subdivision with combinatorial functors.)**.** Recall that there's a functor $V : \Lambda \to \mathrm{Ass}_{act}^{\otimes}$ as constructed in Construction 3.14. Let Free_{C_p} denote the category of finite free C_p-sets and the functor $\mathrm{Free}_{C_p} \to \mathrm{Fin}$ sends objects S to the quotients S/C_p under the free C_p action. There is a natural commutative diagram

$$\begin{tikzcd}
\Lambda^{op} \arrow[r, "\cong"] & \Lambda \arrow[r, "V"] & \mathrm{Ass}_{act}^{\otimes} \\
\Lambda_p^{op} \arrow[u] \arrow[d, "\mathrm{sd}_p"'] \arrow[r, "\cong"] & \Lambda_p \arrow[u] \arrow[d, "\mathrm{sd}_p"'] \arrow[r, "V_p"] & \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes} \arrow[u] \arrow[d] \\
\Lambda^{op} \arrow[r, "\cong"] & \Lambda \arrow[r, "V"] & \mathrm{Ass}_{act}^{\otimes}
\end{tikzcd},$$

where the upper left vertical arrows are the projection $\Lambda_p \to \Lambda = \Lambda_p / (BC_p)$, and he upper right vertical arrow is the projection to $\mathrm{Ass}_{act}^{\otimes}$.The lower right vertical functor is given by the projection to the first factor Free_{C_p} with total orderings on preimages induced by the second factor (i.e.). Finally, the functor V_p sends $[n]_{\Lambda_p}$ to the pair of $V\left(\mathrm{sd}_p\left([n]_{\Lambda_p}\right)\right), V\left([n]_\Lambda\right)$, noting that $V\left(\mathrm{sd}_p\left([n]_{\Lambda_p}\right)\right)$ has a natural C_p -action induced rom the C_p -action on $[n]_{\Lambda_p}$, whose quotient is given by $V\left([n]_\Lambda\right)$

Construction 3.26. We will now consider a variant for when we have BC_p equivariant functors $\Lambda_p^{op} \to S^1\,\mathrm{Top}$. Given such a functor the geometric realisation is an object of $S^1 \times S^1\,\mathrm{Top}$. But by BC_p equivariance we see that the realisation is actually an element of $(S^1 \times S^1)/C_p\,\mathrm{Top}$ which then shows that it is actually an element of $S^1\,\mathrm{Top}$.

Proposition 3.27. *There are natural functors*

$$\mathrm{Fun}^{BC_p}\left(\Lambda_p^{\mathrm{op}}, S^1\,\mathrm{Top}\right) \to S^1\,\mathrm{Top}$$

from the category $\mathrm{Fun}^{BC_p}(\Lambda_p^{op}, S^1\,\mathrm{Top})$ *of* BC_p *-equivariant functors* $\Lambda_p^{op} \to S^1\,\mathrm{Top}$ *to* $S^1\,\mathrm{Top}$*, and*

$$\mathrm{Fun}^{BC_p}\left(\Lambda_p^{\mathrm{op}}, S^1\,\mathrm{Sp}^O\right) \to S^1\mathrm{Sp}^O$$

from the category $\mathrm{Fun}^{BC_p}\left(\Lambda_p^{op}, S^1\mathrm{Sp}^O\right)$ *of* BC_p*-equivariant functors* $\Lambda_p^{\mathrm{op}} \to S^1\mathrm{Sp}^O$ *to* $S^1\mathrm{Sp}^O$ *On the subcategory of functors factoring over a functor* $\Lambda^{op} = \Lambda_p^{op}/BC_p \to \mathrm{Top} \subseteq S^1\,\mathrm{Top}$ *(resp.* $\Lambda^{\mathrm{op}} = \Lambda_p^{\mathrm{op}}/BC_p \to \mathrm{Sp}^O \subseteq S^1\mathrm{Sp}^O$*) , this is given by the usual geometric realization functor discussed in Construction 3.21.*

3.2 Böckstedt's Construction

In this section let $(\mathrm{Sp}^O)^\otimes$ be the Construction 3.5 applied to the symmetric monoidal category Sp^O and let $(\mathrm{Sp}^O)^\otimes_{act}$ denote the Construction 3.9 applied to the category $(\mathrm{Sp}^O)^\otimes$.

Definition 3.28. Let $\mathcal{I}$ be the category of finite (possible empty) sets and injective maps.

Let $A \in \mathrm{Sp}^O$ be an orthogonal spectrum. One gets a natural functor from $\mathcal{I} \to \mathrm{Top}_*$. One does this by identifying $I = \{1, \ldots, i\}$ for $i \geq 0$ and by evaluating A at $A(\mathbb{R}^I)$ as in Eq. 2.1. Here we equip $\mathbb{R}^I$ with the standard metric and a canonical action of $O(i)$. Note that the action of $\Sigma_i \subset O(i)$ acts on $\mathbb{R}^I$ and so this isomorphism is independent of the identification $I = \{1, \ldots, i\}$. This amounts to picking out the i-th space in the sequential model for A.

The category $\mathcal{I}$ comes with a symmetric monoidal structure given by disjoint unions. Moreover it defines a simplicial category by taking products, see [DGM13, Sec. 2.2.2.1] for details, however we will not need them here. The main use of the category $\mathcal{I}$ is that it can be used to show that the Böckstedt's construction is lax monoidal which is not the case when once considers the indexing category to be the category $\mathbb{Z}_{\geq 0}$.

We will use the category $\mathcal{I}$ to define an abstract Böckstedt's construction as a functor from $(\mathrm{Sp}^O)^\otimes_{act}$ to Sp^O. Recall that $(\mathrm{Sp}^O)^\otimes$ is the category constructed from Sp^O as in Def 3.5. If we consider the forgetful functor $p : (\mathrm{Sp}^O)^\otimes \to \mathrm{Fin}_*$ and take the fibre product with the category $\mathrm{Fin} \to \mathrm{Fin}_*$ we obtain a category $(\mathrm{Sp}^O)^\otimes_{act}$.

Definition 3.29. Consider the symmetric monoidal category Sp^O of orthogonal spectra. Bökstedt's construction is the lax symmetric monoidal functor

$$B : \left(\mathrm{Sp}^O\right)^\otimes_{\mathrm{act}} \to \mathrm{Sp}^O$$

that takes a finite family of orthogonal spectra $(X_i)_{i\in I}$ to the orthogonal spectrum whose n -th term is

$$\operatorname*{hocolim}_{(I_i)_{i\in I}\in\mathcal{I}^I} \mathrm{Map}_*\left(\bigwedge_{i\in I} S^{I_i}, S^n \wedge \bigwedge_{i\in I} (X_i)_{I_i}\right)$$

with $O(n)$ -action on S^n, and the natural structure maps.

Here let us explain what this construction is. Here S^{I_i} is the sphere of dimension the cardinality of the set I_i. Since objects in $(\mathrm{Sp}^O)^\otimes_{act}$ are finite sequences of objects Sp^O we see that each X_i in Def. 3.29 in is an object of the category Sp^O. Therefore $(X_i)_{I_i}$ is the value of X_i considered as a functor $\mathcal{I} \to \mathrm{Top}_*$. In fact with this in mind S^{I_i} can be considered as the value of the sphere spectrum $\mathbb{S}$ on the set I_i.

We want to now explain why this construction is actually functorial. To see this note that a morphism in $(\mathrm{Sp}^O)^\otimes$ between objects $(X_i)_{i\in I}$ and $(Y_j)_{j\in J}$ so that there is a map $f : I \to J$ and morphisms $\bigwedge_{i\in f^{-1}(j)} X_i \to Y_j$. This induces a morphism $\bigwedge_{i\in f^{-1}(j)} (X_i)_{I_i} \to (Y_j)_{I_j}$, where $I_j = \coprod_{i\in f^{-1}(j)} I_i$, this follows from noting how the smash product of orthogonal spectra is defined, see [Sch16, Sec. 1]. Finally note that the spheres $\bigwedge_{i\in I} S^{I_i} \cong \bigwedge_{j\in J} S^{I_j}$ upto canonical isomorphism. Finally we get a homomorphism on each level given as

$$\mathrm{Map}_*\left(\bigwedge_{i\in I} S^{I_i}, S^n \wedge \bigwedge_{i\in I} (X_i)_{I_i}\right) \to \mathrm{Map}_*\left(\bigwedge_{i\in I} S^{I_i}, S^n \wedge \bigwedge_{j\in J} (Y_j)_{I_j}\right),$$

and one can pass to the homotopy colimit.

To show that the functor B is lax monoidal we have to show that given orthogonal spectra $(X_i)_{i\in I}$ and $(Y_j)_{j\in J}$, as well as integers m, n, one needs to produce $O(m)\times O(n)$ -equivariant maps from the smash product of

$$\operatorname*{hocolim}_{(I_i)_{i\in I}\in\mathcal{I}^I} \operatorname{Map}_*\left(\bigwedge_{i\in I} S^{I_i}, S^m \wedge \bigwedge_{i\in I} (X_i)_{I_i}\right),$$

and

$$\operatorname*{hocolim}_{(I_j)_{j\in J}\in\mathcal{I}^J}{}_{j\in J} \operatorname{Map}_*\left(\bigwedge_{j\in J} S^{I_j}, S^n \wedge \bigwedge_{j\in J} (Y_j)_{I_j}\right)$$

to the space

$$\operatorname{hocolim}_{(I_k)_{k\in I\sqcup J}\in\mathcal{I}^{I\sqcup J}} \operatorname{Map}_*\left(\bigwedge_{k\in I\sqcup J} S^{I_k}, S^m\wedge S^n \wedge \bigwedge_{k\in I} (X_k)_{I_k} \wedge \bigwedge_{k\in J} (Y_k)_{I_k}\right),$$

which follows from smashing the products.

Another important property of the Böckstedt's construction is its behaviour with respect to the notion of stable equivalences in $(\mathrm{Sp}^O)^{\otimes}_{act}$. Here we say that a map $(X_i)_{i\in I}\to (Y_j)_{j\in J}$ in $\left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}}$ is a stable equivalence if it lies over an equivalence $f: I\to J$ in Fin and the maps $X_i \to Y_{f(i)}$ are stable equivalences for all $i\in I$

Theorem 3.30. *The functor B takes stable equivalences to stable equivalences.*

Proof. See [Shi00, Corollary 4.2.4] and [NS18, Thm. III.4.4.]. □

The main advantage of the Böckstedt's contruction over the cyclic bar construction is the explicit behaviour with respect to base points. To state this precisely and functorial, we will state a many variables functorial version of this construction.

For any finite C_p -set S with $\bar{S} = S/C_p$, consider a family of orthogonal spectra consider the family of orthogonal spectra $(X_{\bar{s}})_{\bar{s}\in\bar{S}}$. Now note that C_p acts on the spectrum $B\left((X_{\bar{s}})_{s\in S}\right)$ by acting on the index sets. Note that in this construction we apply the Böckstedt construction to the family labelled on the lift of $\bar{s}\in\bar{S}$ to S. In particular this means that each such lift of $\bar{s}$ labels the construction applied to the same spectrum, namely $X_{\bar{s}}$. Now we have an action of C_p on the $B\left((X_{\bar{s}})_{s\in S}\right)$ by acting on the indexing set.

Recall that given a finite group G and a category $\mathcal{C}$, the G-equivariant objects in $\mathcal{C}$ can be given as $\mathrm{Func}(BG,\mathcal{C}) := \mathcal{C}^{BG}$ the functor category from G to $\mathcal{C}$. In what follows we will keep this in mind.

Let Free_{C_p} denote the category of finite free C_p -sets. We will consider functors out of the category

$$\mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)^{\otimes}_{\mathrm{act}}\,.$$

In this category objects are pairs of a finite free C_p-set S with quotient $\bar{S} = S/C_p$, together with orthogonal spectra $X_{\bar{s}}$ indexed by $\bar{s}\in\bar{S}$. We consider two functors

$$\mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)^{\otimes}_{\mathrm{act}} \to \mathrm{Sp}^O$$

The first is given by projection to $\left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}}$ and composition with $B: \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}} \to \mathrm{Sp}^O$. This is denoted by B: $\mathrm{Free}_{C_p}\times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}} \to \mathrm{Sp}^O : \left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \mapsto B\left((X_{\bar{s}})_{\bar{s}\in\bar{S}}\right)$

The other functor first applies the functor

$$\mathrm{Free}_{C_p}\times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}} \to \left(\left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}}\right)^{BC_p} : \left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \mapsto \left(S, (X_{\bar{s}})_{s\in S}\right)$$

composes with the Bökstedt construction

$$B: \left(\left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}}\right)^{BC_p} \to \left(\mathrm{Sp}^O\right)^{BC_p} = C_p\mathrm{Sp}^O$$

and then applies

$$\Phi_{\mathcal{U}}^{C_p} : C_p\mathrm{Sp}^O \to \mathrm{Sp}^O$$

for some fixed choice of complete C_p -universe $\mathcal{U}$. We denote this second functor by

$$B_p : \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}} \to \mathrm{Sp}^O : \left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \mapsto \Phi_{\mathcal{U}}^{C_p} B\left((X_{\bar{s}})_{s\in S}\right)$$

Construction 3.31. There is a natural transformation

$$B_p \to B : \Phi_{\mathcal{U}}^{C_p} B\left((X_{\bar{s}})_{s\in S}\right) \to B\left((X_{\bar{s}})_{\bar{s}\in\bar{S}}\right)$$

of functors

$$\mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}} \to \mathrm{Sp}^O$$

To see this note that note that the space on the left side $\Phi_{\mathcal{U}}^{C_p}(B(X_{\bar{s}})_{s\in S})$, is given by

$$\operatorname*{hocolim}_{V\in\mathcal{U}|V^Cp=0}\left(\operatorname*{hocolim}_{(I_s)\in\mathcal{I}^S}\mathrm{Map}_*\left(\bigwedge_{s\in S} S^{I_s}, S^n\wedge S^V\wedge\bigwedge_{s\in S} X_{\bar{s},I_s}\right)\right)^{C_p}.$$

For any $V\in\mathcal{U}$ with $V^{C_p}=0$, we have to construct a map from

$$\left(\operatorname*{hocolim}_{(I_s)\in\mathcal{I}^S}\mathrm{Map}_*\left(\bigwedge_{s\in S} S^{I_s}, S^n\wedge S^V\wedge\bigwedge_{s\in S} X_{\bar{s},I_s}\right)\right)^{C_p}$$

to

$$\operatorname*{hocolim}_{(I_{\bar{s}})\in\mathcal{I}^{\bar{S}}}\mathrm{Map}_*\left(\bigwedge_{\bar{s}\in\bar{S}} S^{I_{\bar{s}}}, S^n\wedge\bigwedge_{\bar{s}\in\bar{S}} X_{\bar{s},I_{\bar{s}}}\right).$$

We refer the reader to [NS18, Construction. III 4.6] to see the full construction. The idea is to examine the fixed points under the C_p action and note that maps have to map fixed points to fixed points.

Finally we can state the main theorem of this section.

Theorem 3.32. *For any*

$$\left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \in \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)^{\otimes}_{\mathrm{act}},$$

the natural map

$$B_p\left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \to B\left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) : \Phi_{\mathcal{U}}^{C_p} B\left((X_{\bar{s}})_{s\in S}\right) \to B\left((X_{\bar{s}})_{\bar{s}\in\bar{S}}\right)$$

of Construction 3.31 is a stable equivalence.

Proof. See [HM97, Prop. 2.5] and [NS18, Thm. III.4.7] for a proof. ∎

3.3 Topological Hochschild Homology

We would now like to describe a method to define a topological analogue of Hochschild homology and a topological analogue of cyclic homology. For any ordinary associative ring, A one has the Eilenberg-Maclane spectrum functor which associates to it an associative algebra object (as in Def. 3.7) HA in the category of orthogonal spectra as in Example 2.2 (2), one may work more generally with associative algebra objects in Sp^O.

Since in Sp^O, the initial ring spectrum is $\mathbb{S}$ (Example 2.2 (1)), one may, given an associative ring spectrum R, hope to construct the cyclic bar complex for R.

$$\ldots \Rrightarrow R\wedge_{\mathbb{S}} R\wedge_{\mathbb{S}} R \Rrightarrow R\wedge_{\mathbb{S}} R \rightrightarrows R$$

While this will work, it was not clear until recently (see [ABG+14]) how to construct the cyclotomic structure on the resulting spectrum in the case of orthogonal spectra. We will instead proceed by applying the Böckstedt construction to R, which is essentially a fibrant replacement of the cyclic bar construction.

We will freely use the language and notation of Section. 3.1.

Definition 3.33. Let R be an orthogonal ring spectrum.

1. For any pointed space X, define a pointed space with S^1 -action $\mathrm{THH}(R;X)$ as the geometric realization of the cyclic space sending $[k]_\Lambda \in \Lambda^{\mathrm{op}}$, $k \geq 1$, to

$$B(R,\ldots,R) = \operatorname*{hocolim}_{(I_1,\ldots,I_k)\in\mathcal{I}^k} \mathrm{Map}_*\left(S^{I_1}\wedge\ldots\wedge S^{I_k}, X\wedge R_{I_1}\wedge\ldots\wedge R_{I_k}\right)$$

where Map_* is the based mapping space.

2. The orthogonal S^1 -spectrum $\mathrm{THH}(R)$ has the m -th term given *by* $\mathrm{THH}\left(R;S^m\right)$ with the $O(m)$-action induced by the action on S^m. The structure maps σ_m are the natural maps

$$\mathrm{THH}\left(R;S^m\right)\wedge S^1 \to \mathrm{THH}\left(R;S^m\wedge S^1\right) = \mathrm{THH}\left(R;S^{m+1}\right)$$

Remark 3.34. We note that the construction first gives us a cyclic spectrum (see Construction 3.12), $\mathrm{THH}_\bullet(R)$. One can then by Construction 3.19 take geometric realisations to get the S^1-spetrum $\mathrm{THH}(R)$. Thus we may as well describe the construction functorially as follows. There is a natural map

$$\Lambda^{op} \to \mathrm{Ass}^{\otimes}_{act}$$

given in Construction 3.14. Then as described in Construction 3.7 we get an associative algebra object

$$R^{\otimes} : \mathrm{Ass}^{\otimes} \to (\mathrm{Sp}^O)^{\otimes}$$

which is exactly the orthgonal ring spectrum R (we restrict this functor to the active subcategories). The Böckstedt Construction,

$$B : (\mathrm{Sp}^O)^{\otimes}_{act} \to \mathrm{Sp}^O$$

of Def. 3.29 then composes with it to land in Sp^O.

We now want to describe a natural cyclotomic structure on the object $\mathrm{THH}(R)$. Let us recall what this construction is going to be. We need to fix a complete S^1-univers $\mathcal{U}$ with respect to which we have fixed point functors $\Phi^{C_p}_{\mathcal{U}}$ for all $p \geq 1$. We want to show that these functors satsify a natural $\mathcal{F}$ equivalence, (where $\mathcal{F}$ is the class of all finite subgroups of S^1) for all $p \geq 1$

$$\Phi_p : \Phi^{C_p}_{\mathcal{U}}(\mathrm{THH}(R)) \to \mathrm{THH}(R)$$

and satisfy the coherence conditions in Def. 2.17.

Our strategy in this construction would be to induce an equivalence of S^1-spectra by using simplicial subdivision. We will hinge on the very important Theorem 3.32 relying crucially on the functor B_p we constructed in Section 3.2 and its equivalence with the Böckstedt's construction in Def. 3.29. For this we need to bootstrap the functor $\Phi^{C_p}_{\mathcal{U}}$ to come out of the category $C_p\mathrm{Sp}^O$, the category of orthogonal spectra with a C_p action. This can always be done by choosing a complete S^1-universe cleverly.

Recall that any S^1-complete universe gives a complete C_p universe for all finite subgroups of S^1 by the discussion after Def.2.12.

We fix a complete S^1-universe as follows

$$\mathcal{U} = \bigoplus_{k\in\mathbb{Z}, i\geq 1} \mathbb{C}_{k,i}$$

as in the definition of orthogonal cyclotomic spectra, Def. 2.17. Here S^1 acts on $\mathbb{C}_{k,i}$ via the k-th power of the embedding $\mathbb{T} \hookrightarrow \mathbb{C}^\times$.

We want to consider the universe $\mathcal{U}$ as a complete C_p universe as in the discussed and we consider $\mathrm{THH}(R)$ as having a $C_p \subset S^1$ action instead of the full S^1 action.

In this case the functors $\Phi^{C_p}_{\mathcal{U}}$ will a priori take value in Sp^O instead of taking value in $S^1\mathrm{Sp}^O$. However as it turns out for any C_p spectrum X, $\Phi^{C_p}_{\mathcal{U}}X$ will instead land in $S^1\mathrm{Sp}^O$, (here compare with Lemma. 2.14.) The reason is that in the formula for $\Phi^{C_p}_{\mathcal{U}}X)$ one has

$$\Phi^{C_p}_{\mathcal{U}}(X)_n = \operatorname*{hocolim}_{V\in\mathcal{U}|V^{C_p}=0} X\left(\mathbb{R}^n \oplus V\right)^{C_p},$$

and V still has a residual S^1 action. This is a subtlety and happens exactly because of our choice of the universe, $\mathcal{U}$.

The resulting functor

$$\Phi_{\mathcal{U}}^{C_p} : C_p\mathrm{Sp}^O \to S^1\mathrm{Sp}^O \tag{3.1}$$

is BC_p equivariant. Here BC_p action on the category $C_p\mathrm{Sp}^O$ comes from the action on itself and on $S^1\mathrm{Sp}^O$ the action comes from the restriction of the S^1-action.

Now we look at $\mathrm{THH}(R)$ as a C_p-spectrum via restriction of the S^1 action. Now

$$\Phi_{\mathcal{U}}^{C_p} \mathrm{THH}(R)_n = \underset{V\in\mathcal{U}|V^{C_p}=0}{\mathrm{hocolim}} \mathrm{THH}\left(R; S^n \wedge S^V\right)^{C_p},$$

where the C_p action on $\mathrm{THH}(R; S^n \wedge S^V)$ is given by the diagonal action combining the action on the representation V and the action of $C_p \subseteq \mathbb{T}$ on THH. We can now reinterpret this formula using simplicial subdivision (Construction 3.22).

The spectrum $\mathrm{THH}(R)$ can also be interpreted as the geometric realisation of the Λ_p^{op} orthogonal spectrum via the simplicial subdivision functor $\mathrm{sd}_p : \Lambda_p^{op} \to \Lambda^{op}$ (see Remark 3.24.) 2). This Λ_p^{op} -orthogonal spectrum is equivalently given by the composition of

$$\Lambda_p^{\mathrm{op}} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes}$$

with

$$\mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)_{act}^{\otimes}$$

and the Böckstedt construction

$$B : ((\mathrm{Sp}^O)_{\mathrm{act}}^{\otimes})^{BC_p} \to (\mathrm{Sp}^O)^{BC_p} = C_p\mathrm{Sp}^O.$$

Thus overall we get a functor Thus overall we get the following proposition

Proposition 3.35. *There is a natural map of orthogonal spectra from $\Phi_{\mathcal{U}}^{C_p}\mathrm{THH}(R)$ to the geometric realization of the Λ_p^{op} -orthogonal spectrum given as the composition of*

$$\Lambda_p^{\mathrm{op}} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{\mathrm{act}}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)_{\mathrm{act}}^{\otimes}$$

with the functor

$$B_p : \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)_{\mathrm{act}}^{\otimes} \to \mathrm{Sp}^O : \left(S, (X_{\bar{s}})_{\bar{s}\in\bar{S}}\right) \mapsto \Phi_{\mathcal{U}}^{C_p} B\left((X_{\bar{s}})_{s\in S}\right)$$

This map is a homeomorphism of orthogonal spectra.

Proof. The canonical map

$$|\Phi_{\mathcal{U}}^{C_p} \mathrm{sd}_p \mathrm{THH}(R)| \to \Phi_{\mathcal{U}}^{C_p} |\mathrm{sd}_p \mathrm{THH}(R)| \cong \Phi_{\mathcal{U}}^{C_p} |\mathrm{THH}(R)|$$

is an isomorphism. See [NS18, III 5.4]. ☐

However this construction is not quite right because it does not keep track of the S^1 action on $\Phi_{\mathcal{U}}^{C_p} \mathrm{THH}(R)$ coming in through the fact that this is an Λ_p^{op} spectrum. To correct this we have to use the refinement discussed in Eq. 3.1 in the last step. In this case the functor $B_p^{S^1}$ so defined is the same as the functor B_p constructed in Section 3.2 except we now refine the functor

$$\Phi_{\mathcal{U}}^{C_p} : C_p\mathrm{Sp}^O \to S^1\mathrm{Sp}^O.$$

More precisely, using this refinement, we get a similar refinement

$$B_p^{S^1} : \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)_{\mathrm{act}}^{\otimes} \to S^1\mathrm{Sp}^O.$$

This functor is again BC_p equivariant because the functor $\Phi_{\mathcal{U}}^{C_p}$ is.

Let us now recall what BC_p equivariant functors from $\Lambda_p^{op} \to S^1\,\mathrm{Top}$ mean. These are essentially functors which factor over the the map $\Lambda_p^{op} \to \Lambda_p^{op}/BC_p \cong \Lambda^{op}$. Thus the image will now have trivial C_p action. Once one takes the geometric realisation, a priori these are objects of $S^1 \times S^1\,\mathrm{Top}$. However because of the trivial action C_p one gets an action of $(S^1 \times S^1)/C_p$. Upon restricting on the diagonal one gets an S^1 action. This is the content of Prop. 3.27.

In particular, the BC_p -equivariant functor

$$\Lambda_p^{\mathrm{op}} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{\mathrm{act}}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)_{\mathrm{act}}^{\otimes} \xrightarrow{B_p^{\mathbb{T}}} \mathbb{T}\mathrm{Sp}^O$$

gives rise to an orthogonal spectrum with a continuous $(\mathbb{T} \times \mathbb{T})/C_p$ -action. Restricting to the diagonal $\mathbb{T}/C_p \cong \mathbb{T}$ -action, we get an orthogonal spectrum with T-action, (cf.Prop 3.27).

The following proposition follows in the same way as Prop. 3.35.

Proposition 3.36. *The S^1 -equivariant orthogonal spectrum $\Phi_{\mathcal{U}}^{C_p}\,\mathrm{THH}(R) \in S^1\,\mathrm{Sp}^O$ has a natural map to the object of $S^1\mathrm{Sp}^O$ corresponding to the* BC_p *-equivariant functor*

$$\Lambda_p^{\mathrm{op}} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{\mathrm{act}}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \left(\mathrm{Sp}^O\right)_{\mathrm{act}}^{\otimes} \xrightarrow{B_p^{S^1}} \mathrm{TSp}^O$$

under the functor of Prop 3.27. This map is a homeomorphism of S^1-orthogonal spectra.

In this picture one may also recover $\mathrm{THH}(R)$ itself.

Proposition 3.37. *The S^1-equivariant orthogonal spectrum $\mathrm{THH}(R) \in S^1\mathrm{Sp}^O$ is given by the object of $S^1\mathrm{Sp}^O$ corresponding to the BC_p-equivariant functor*

$$\Lambda_p^{op} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)_{act}^{\otimes} \xrightarrow{B} \mathrm{Sp}^O \subset S^1\mathrm{Sp}^O$$

where B is as in Construction 3.31 and in the final inclusion we equip any orthogonal spectrum with a trivial S^1-action.

Proof. Follows from equivalence of realisation of simplicial subdivision. □

Finally Construction 3.31 induces an equivalence of BC_p equivariant functors

$$\Lambda_p^{op} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)_{act}^{\otimes} \xrightarrow{B_p^{S^1}} S^1\mathrm{Sp}^O,$$

and

$$\Lambda_p^{op} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} \mathrm{Ass}_{act}^{\otimes} \to \mathrm{Free}_{C_p} \times_{\mathrm{Fin}} (\mathrm{Sp}^O)_{act}^{\otimes} \xrightarrow{B} \mathrm{Sp}^O \subset S^1\mathrm{Sp}^O.$$

Applying Prop. 3.27, the maps induces a transformation

$$\Phi_p : \Phi_{\mathcal{U}}^{C_p}\,\mathrm{THH}(R) \to \mathrm{THH}(R)$$

of objects of $S^1\mathrm{Sp}^O$.

By Theorem 3.32, these maps are equivalences of the underlying orthogonal spectra. Moreover, it is easy to see that these maps sit in commutative diagrams for all $p, q \geq 1$

$$\begin{tikzcd}
\Phi_{\mathcal{U}}^{C_p}(\Phi_{\mathcal{U}}^{C_q} X) \arrow[r, "\cong"] \arrow[d, "\Phi_{\mathcal{U}}^{C_p}(\Phi_q)"'] & \Phi_{\mathcal{U}}^{C_{pq}} X \arrow[d, "\Phi_{pq}"] \\
\Phi_{\mathcal{U}}^{C_p} X \arrow[r, "\phi_q"] & X
\end{tikzcd},$$

as in Definition 2.17. Since the maps $\Phi_{\mathcal{U}}^{C_p}(\Phi_q)$ are equivalences of underlying spectra, it is easy to see that they are $\mathcal{F}$ equivalences.

3.4 THH and TC of a ring

We can now define the topological Hochschild homology (THH), Def. 3.33 and topological cyclic homology (TC), Def. 2.18 of an ordinary associative ring.

Recall that every ordinary associative ring A gives rise to an associative rings spectrum HA in the category of orthogonal spectra, Sp^O, as in Example 2.2 (2).

Definition 3.38. Let A be an ordinary ring. Then

1. The topological Hochschild homology of A, $\mathrm{THH}(A)$ is the topological Hochschild homology of the associative ring spectrum HA.
2. The topological cyclic homology of A, $\mathrm{TC}(A)$ is the topological cyclic homology of the cyclotomic spectrum $\mathrm{THH}(A)$.

4 Calculations

The aim of this section is to give a calculation of $\mathrm{TC}(\mathbb{F}_p)$, for p a prime. Before beginning we invite the reader to compare with the calculation carried out in [NS18, Section. IV.4] for a finer calculation than the one we do now using modern techniques. Their calculation uses a new definition of TC which agrees with ours on bounded below spectra, and as is pertinent to our case, with Eilenberg-Maclane spectra. Their methods are particularly amenable to calculations, for example see [Spe18].

In this section we give the calculation of $\mathrm{TC}(\mathbb{F}_p)$ our main sources are the original work of Hesselholt-Madsen in [HM97, Section 5] and Madsen's survey [Mad94]. Other important sources are Böckstedt and Madsen's calculations of $\mathrm{TC}(\mathbb{Z}_p)$ in [BM94], where the spectral sequences we will be using where set up, and the Greenless-May foundations in [GM95]. We will also use Böckstedt and Breen's calculation of

We will set up some conventions. We will implicitly remain in the setting of [LMSM86], and assume that all our FSP's give the correct notion of Eilenberg-Maclane spectra. We will use notation as in [HM97].

4.1 Calculations of $\mathrm{TC}(\mathbb{F}_p)$

4.1.1 Preliminaries

Let $\mathcal{U}$ be a complete C universe where C is some finite group. Let $C\mathcal{SU}$ be the category of C-spectra indexed on the $\mathcal{U}$. There is an inclusion functor for the C trivial universe

$$j : \mathcal{U}^C \to \mathcal{U}.$$

Let T be a C-spectrum in the sense of [LMSM86]. We can pull back this spectrum the the C trivial universe $\mathcal{U}^C$ and this spectrum is denoted j^*T.

Then following Greenless-May (cf. [GM95]) we define

$$\begin{aligned} T_{hC} &= j^*T \wedge_C EC_+ \text{ (homotopy orbit)} \\ T^{hC} &= F(EC_+, T)^C \text{ (homotopy fixed points)} \\ \hat{\mathbb{H}}(C;T) &= [\tilde{E}C \wedge F(EC_+, T)]^C \text{ (Tate spectrum).} \end{aligned}$$

By smashing the cofibration sequence of C-spaces

$$EC_+ \to S^0 \to \tilde{E}C \tag{4.1}$$

with the function spectrum $F(EC_+, T) \in C\mathcal{SU}$ and taking C-fixed points one gets the 'norm cofibration' sequence' of [GM95],

$$T_{hC} \xrightarrow{N^h} T^{hC} \xrightarrow{R^h} \hat{\mathbb{H}}(C;T). \tag{4.2}$$

In fact if we assume $T \in S^1\mathcal{SU}$ and let C be a cyclic p-subgroup. In [HM97, Prop. 2.1] the authors identify $[\tilde{E}C \wedge T]^C$ with $(\Phi^{C_p}T)^{C/C_p}$. Therefore one can smash the inclusion

$$\gamma : T \to F(EG_+, T)$$

with $\tilde{E}C$ to obtain a C/C_p equivariant map $\hat{\gamma} : \Phi^{C_p}T \to \hat{\mathbb{H}}(C_p; T)$ and taking fixed sets and using that $\hat{\mathbb{H}}(C_p, T)^{C/C_p} = \hat{\mathbb{H}}(C; T)$ we obtain a cofibration diagram

$$\begin{tikzcd} T_{hC} \arrow[r, "N"] \arrow[d, "="'] & T^C \arrow[r] \arrow[d, "\Gamma_C"] & (\Phi^{C_p}T)^{C/C_p} \arrow[d, "\hat{\Gamma}_C"] \\ T_{hC} \arrow[r, "N^h"] & T^{hC} \arrow[r, "R^h"] & \mathbb{H}(\hat{C}; T) \end{tikzcd} \tag{4.3}$$

For a cyclotomic spectrum this reduces to the equivalence

Proposition 4.1. *[BM94] For a p-cyclotomic spectrum T there is a commutative diagram*

$$\begin{tikzcd} T_{hC} \arrow[r, "N"] \arrow[d, "="'] & T^{C_{p^n}} \arrow[r, "R"] \arrow[d, "\Gamma_C"] & T^{C_{p^{n-1}}} \arrow[d, "\hat{\Gamma}_C"] \\ T_{hC} \arrow[r, "N^h"] & T^{hC} \arrow[r, "R^h"] & \mathbb{H}(\hat{C}; T) \end{tikzcd} \tag{4.4}$$

In our case the spectrum we have to deal with is called the topological Hocschild homology spectrum. In our case the cofibration sequence can be recorded as

Proposition 4.2. *There is a homotopy commutative diagram of cofibrations (of non-equivariant spectra)*

$$\begin{tikzcd} \mathrm{THH}(L)_{hC_{p^n}} \arrow[r, "N"] \arrow[d] & \mathrm{THH}(L)^{C_{p^n}} \arrow[r, "R"] \arrow[d, "\Gamma"] & \mathrm{THH}(L)^{C_{p^{n-1}}} \arrow[d, "\hat{\Gamma}"] \\ \mathrm{THH}(L)_{hC_{p^n}} \arrow[r] & \mathrm{THH}(L)^{hC_{p^n}} \arrow[r] & \hat{\mathbb{H}}(C_{p^n}, \mathrm{THH}(L)) \end{tikzcd} \tag{4.5}$$

One may attempt to get information about the homotopy groups of the objects in the Norm cofibration sequence in Eq 4.2 via spectral sequences.

Let M be a coefficient group (usually $M = \mathbb{Z}_p$ or $M = \mathbb{F}_p$). to ensure convergence we will have to assume that each $\pi_*(T(L), M)$ is a finitely generated $\mathbb{Z}_p$ module in each degree.

The spectral sequences which will help us compute were set up in [BM94, Sec. 2] and in [GM95, Sec. 10].

The spectral sequences in our case are given by

$$E^2_{s,t}\left(THH(L)_{hC_{p^n}}; M\right) = H_s\left(C_{p^n}; \pi_t(THH(L); M)\right) \Rightarrow \pi_{s+t}\left(THH(L)_{hC_{p^n}}; M\right) \tag{4.6}$$

$$E^2_{-s,t}\left(THH(L)^{hC_{p^n}}; M\right) = H^s\left(C_{p^n}; \pi_t(THH(L); M)\right) \Rightarrow \pi_{t-s}\left(THH(L)^{hC_{p^n}}; M\right) \tag{4.7}$$

$$E^2_{-s,t}\left(\hat{\mathbb{H}}\left(C_{p^n}, THH(L)\right); M\right) = \hat{H}^s\left(C_{p^n}; \pi_t(THH(L); M)\right) \Rightarrow \pi_{t-s}\left(\hat{\mathbb{H}}\left(C_{p^n}; THH(L)\right); M\right) \tag{4.8}$$

The spectral sequences are concentrated in the upper half plane and the differentials take $E^r_{s,t}$ to $E^r_{s-r,t+r-1}$ and for commutative FSP L the last two spectral sequences have ring structure when M is a p-adic ring with p odd. Since the C_{p^n}-action comes from an S^1-action, $\pi_*(T(L); M)$ has trivial C_{p^n} action. For p odd:

$$\begin{aligned} E^2\left(THH(L)^{hC_{p^n}}; \mathbb{F}_p\right) &= E\{u_n\} \otimes S\{t\} \otimes \pi_*\left(THH(L); \mathbb{F}_p\right) \\ E^2\left(\hat{\mathbb{H}}\left(C_{p^n}, THH(L)\right); \mathbb{F}_p\right) &= E\{u_n\} \otimes S\left\{t, t^{-1}\right\} \otimes \pi_*\left(THH(L); \mathbb{F}_p\right) \end{aligned} \tag{4.9}$$

with $\deg(u) = (-1, 0), \deg(t) = (-2, 0)$ and $\pi_t(THH(L); \mathbb{F}_p)$ sitting in degree $(0, t)$.

In the above H_s, H^s and $\hat{H}^s$ denotes group homology, group cohomology and group Tate cohomology. They are related by the formulas:

$$\hat{H}^{-s}(G;A) = \begin{cases} H^{-s}(G;A), & s \leq 0 \\ H_{s-1}(G;A), & s > -1 \\ \ker\left(\text{ Norm } : H_0(G;A) \to H^0(G;A)\right), & s = -1 \\ \operatorname{coker}\left(\text{ Norm } : H_0(G;A) \to H^0(G;A)\right), & s = 0 \end{cases}$$

When $G = C_{p^n}$ and $pA = 0$ then Norm $= 0$ so we see that

$$E^2_{-s,t}\left(\hat{\mathbb{H}}\left(C_{p^n}; THH(L)\right); \mathbb{F}_p\right) = \begin{cases} E^2_{-s,t}\left(THH(L)^{hC_{p^n}}; \mathbb{F}_p\right), & s \geq 0 \\ E^2_{-s-1,t}\left(THH(L)_{hC_{p^n}}; \mathbb{F}_p\right), & s < 0 \end{cases}$$

The d^2-differential in the spectral sequences is connecteed to the action

$$A : S^1_+ \wedge \mathrm{THH}(L) \to \mathrm{THH}(L)$$

as follows. The stable homotopy $\pi_1^S(S^1_+) \cong \pi_4(\sigma^3(S^1_+)) \xrightarrow{\cong} \pi_4(S^2) \oplus \pi_4(S^3)$ is $\mathbb{Z} \oplus \mathbb{Z}/2$ generated by the $\sigma = id$ and the Hopf map η. Thus we get operators

$$[S^1], \eta : \pi_i(\mathrm{THH}(L)) \to \pi_{i+1}(S^1_+ \wedge \mathrm{THH}(L)) \xrightarrow{A_*} \pi_{i+1}\,\mathrm{THH}(L)$$

where the first map is exterior product with σ and ηrespectively. There are induced operations

$$\hat{H}^s\left(C_{p^n}; \pi_t(\mathrm{THH}(L); M)\right) \to \hat{H}^s\left(C_{p^n}; \pi_{t+1}(\mathrm{THH}(L); M)\right)$$

which we can compose with the periodicity isomorphism

$$\hat{H}^s\left(C_{p^n}; \pi_{t+1}(\mathrm{THH}(L); M)\right) \xrightarrow{\simeq} \hat{H}^{s+2}\left(C_{p^n}; \pi_{t+1}(\mathrm{THH}(L); M)\right)$$

to get maps $[S^1]_{\#}.\eta_{\#}$.

Proposition 4.3. *In the spectral sequence $E^r_{*,*}\left(\hat{\mathbb{H}}\left(C_{p^n}, \mathrm{THH}(L)\right); M\right)$ the d^2 -differential*

$$d^2 : \hat{H}^s\left(C_{p^n}; \pi_t(\mathrm{THH}(L); M)\right) \to \hat{H}^{s+2}\left(C_{p^n}; \pi_t(\mathrm{THH}(L); M)\right)$$

is equal to $\left[S^1\right]_{\#}$, provided η acts trivially on $\pi_*(\mathrm{THH}(L); M)$.

Proof. See [Hes96]. □

Another factor that we have to consider in our computation is the homotopical behaviour of the map

$$\hat{\Gamma}_n : \mathrm{THH}(L)^{C_{p^{n-1}}} \to \hat{\mathbb{H}}(C_{p^n}, \mathrm{THH}(L))$$

of the cofibration sequence in 4.5. We will need the following theorem of Tsaldis.

Theorem 4.4. *[Tsa98] Suppose*

$$\pi_i(\hat{\Gamma}_1) : \pi_i\left(\mathrm{THH}(L); \mathbb{F}_p\right) \to \pi_i\left(\hat{\mathbb{H}}\left(C_p, \mathrm{THH}(L)\right); \mathbb{F}_p\right)$$

is an isomorphism for $i \geq i_0$ then the same is true for $\pi_i(\hat{\Gamma}_n)$ for all $n \geq 1$.

4.1.2 Calculations

For any given ring A the associated $\mathrm{THH}(A)$ can always be described as the realisation of a simplicial abelian group (cf. [Mad94, Sec. 3.2]), and then in particular its homotopy groups completely determine the homotopy type of of $\mathrm{THH}(A)$. More precisely it is non-canonically equivalent to a wedge of Eilenberg-Maclane spectra.

The simplicial filtration of $\mathrm{THH}(A)$ gives rise to a spectral sequence

$$E^2(A)=\mathrm{HH}_*(H_*(A);\mathbb{F}_p)\Rightarrow \pi_*(\mathrm{THH}(A);\mathbb{F}_p) \tag{4.10}$$

This calculation was used by [Böc85b]. A modern version of this calculation using the language developed by Grothendieck and Berthelot in [Ber74] is given in the survey [HN19]. In particular Hesselholt and Nikolaus present a derivation of the spectral sequence using derived descent on a particular stack.

The 0-skeleton of $\mathrm{THH}(A)$ is the Eilenberg-Maclane spectrum HA and one may use the S^1 action to get the map

$$\sigma : S^1_+\wedge HA\to S^1_+\wedge \mathrm{THH}(A)\to \mathrm{THH}(A) \tag{4.11}$$

For $A=\mathbb{F}_p$ we have $\tau_0\in\pi_1(H\mathbb{F}_p;\mathbb{F}_p)$ and can consider $\sigma_*\left(\left[S^1\right]\wedge\tau_0\right)\in\pi_2\left(\mathrm{THH}\left(\mathbb{F}_p\right);\mathbb{F}_p\right)$, where $\left[S^1\right]\in\pi_1^S\left(S^1_+\right)$ was defined in the previous section.

Theorem 4.5. *[Böc85b][Bre78] The reduction*

$$\mathrm{red}_p:\pi_2\mathrm{TH}\left(\mathbb{F}_p\right)\to\pi_2\left(\mathrm{TH}\left(\mathbb{F}_p\right);\mathbb{F}_p\right)$$

is an isomorphism, and

$$\pi_*\mathrm{TH}\left(\mathbb{F}_p\right)=S_{\mathbb{F}_p}\{\sigma\}$$

the polynomial algebra on σ of degree 2 with $\mathrm{red}_p(\sigma)=\sigma_*\left(\left[S^1\right]\wedge\tau_0\right)$

Combined with 4.9 we can explicate the E^2 -terms of the spectral sequence

$$\hat{E}^r\left(C_{p^n};M\right)=E^r\left(\hat{\mathbb{H}}\left(C_{p^n},T\left(\mathbb{F}_p\right)\right);M\right)$$

for $M=\mathbb{F}_p,\ \mathbb{Z}_p$ to be

$$\begin{aligned}\hat{E}^2\left(C_{p^n};\mathbb{F}_p\right)&=E_{\mathbb{F}_p}\left\{u_n\right\}\otimes S_{\mathbb{F}_p}\left\{t,t^{-1}\right\}\otimes E_{\mathbb{F}_p}\left\{e_1\right\}\otimes S_{\mathbb{F}_p}\{\sigma\}\\ \hat{E}^2\left(C_{p^n};\mathbb{Z}_p\right)&=E_{\mathbb{F}_p}\left\{u_n\right\}\otimes S_{\mathbb{F}_p}\left\{t,t^{-1}\right\}\otimes S_{\mathbb{F}_p}\{\sigma\}\end{aligned} \tag{4.12}$$

except if $p=2$ and $n=1$ where the first two terms are replaced by $S\left\{u_1,u_1^{-1}\right\}$. The mod p Bockstein operator maps $e_1\sigma^l$ to σ^l for $l\geq 0$. For p odd, $\hat{E}^r\left(C_{p^n};\mathbb{F}_p\right)$ is a spectral sequence of algebras. If $p=2$ there is the usual trouble with products in $\pi_*\left(T;\mathbb{F}_2\right)$ but in all cases, $\hat{E}^r\left(C_{p^n};\mathbb{F}_p\right)$ is an algebra over $\hat{E}^r\left(C_{p^n};\mathbb{Z}_p\right)$

Lemma 4.6. *The non-zero differentials in $\hat{E}^r\left(C_{p^n};\mathbb{F}_p\right)$ are generated from $d^2e_1=t\sigma$ in the module structure over $\hat{E}^r\left(C_{p^n};\mathbb{Z}_p\right)$. In particular*

$$\begin{aligned}&\pi_*\left(\hat{\mathbb{H}}\left(C_{p^n},\mathrm{TH}\left(\mathbb{F}_p\right)\right);\mathbb{F}_p\right)\cong E_{F_p}\left\{u_n\right\}\otimes S_{F_p}\left\{t,t^{-1}\right\},\quad p\ \textit{odd or}\ n>1\\ &\pi_*\left(\hat{\mathbb{H}}\left(C_2,\mathrm{TH}\left(\mathbb{F}_2\right)\right);\mathbb{F}_2\right)\cong S_{\mathbb{F}_2}\left\{u_1,u_1^{-1}\right\}\end{aligned} \tag{4.13}$$

Proof. Since $e_1=\sigma_*\left(\tau_0\right),\tau_0\in\pi_1\left(H\mathbb{F}_p;\mathbb{F}_p\right)$ and $\mathrm{red}_p(\sigma)=\sigma_*\left(\left[S^1\right]\wedge\tau_0\right)$ we have in the notation of proposition 4.1 .14

$$\left[S^1\right]_\#\left(e_1\right)=\sigma,\quad \left[S^1\right]_\#(1)=0$$

and hence $\left[S^1\right]_\#\left(e_1\sigma^l\right)=\sigma^{l+1}$. The d^2 -differential then follow from (4.1.14) and a routine cohomology calculation gives

$$\hat{E}^3\left(C_{p^n};\mathbb{F}_p\right)=E_{F_p}\left\{u_n\right\}\otimes S_{\mathbb{F}_p}\left\{t,t^{-1}\right\}$$

(with $u_1^2=t$ if $p=2$ and $n=1$). For degree reasons there can be no further differentials. For p odd (and $p=2,n=1$) this is a free commutative algebra in the graded sense, and the stated value of the mod p homotopy is immediate. If $p-2$ and $n>1$ one uses that the mod p Bockstein on u_n is trivial. □

For $n = 1$, the mod p Bockstein relation $\beta(u_1) = t$ gives that

$$\pi_* \hat{\mathbb{H}}(C_p, TH(\mathbb{F}_p)) = S_{\mathbb{F}_p}\left\{t, t^{-1}\right\}$$

(with $t = u_1^2$ if $p = 2$).

We would now like to use Tsadalis Theorem (Theorem 4.4) in order to solve the homotopy limit problem.

Lemma 4.7. *The homomorphism*

$$\pi_i(\hat{\Gamma}_1; \mathbb{F}_p) : \pi_i(\mathrm{TH}(\mathbb{F}_p); \mathbb{F}_p) \to \pi_i\left(\hat{\mathbb{H}}(C_p, \mathrm{TH}(\mathbb{F}_p)); \mathbb{F}_p\right)$$

is an isomorphism when $i \geq 0$

Proof. See [Mad94, Lemma. 4.2.6] □

Now Lemma 4.7 shows that the morphism

$$\hat{\Gamma}_n : \mathrm{THH}(A)^{C_{p^n}} \to \hat{\mathbb{H}}\left(C_{p^{n+1}}, TH(\mathbb{F}_p)\right)$$

is (-1)-connected. Furtermore using the fact that $\mathrm{THH}(\mathbb{F}_p)$ is p-complete, one can inductively prove that $\mathrm{THH}(\mathbb{F}_p)^{C_{p^n}}$ is p-complete by inductive use of the top row of the diagram 4.5. Thus

$$\pi_*(\mathrm{THH}(\mathbb{F}_p))^{C_{p^n}} = \pi_*(\mathrm{THH}(\mathbb{F}_p)^{C_{p^n}}, \mathbb{Z}_p) \tag{4.14}$$

One can now use the algebra structure of the spectral sequence

$$E^r(\hat{\mathbb{H}}(C_{p^n}, T(\mathbb{F}_p); \mathbb{F}_p)$$

over

$$E^r(\hat{\mathbb{H}}(C_{p^n}, T(\mathbb{F}_p); \mathbb{Z}_p)$$

to conclude the following proposition.

Proposition 4.8. *For* $n \geq 1$

$$\pi_* \mathrm{THH}(\mathbb{F}_p)^{C_{p^n}} = S_{\mathbb{Z}/p^{n+1}}\{\sigma_n\}$$

with $\deg \sigma_n = 2$. *Moreover,* $F(\sigma_n) = \sigma_{n-1}$ *and* $R(\sigma_n) = \lambda_n p \sigma_{n-1}$ *with* $\lambda_n \in \mathbb{Z}/p^n$ *a unit.*

Proof. See [Mad94, Prop. 4.2.7] □

Corollary 4.9.

$$\mathrm{TC}(\mathbb{F}_p) \simeq H\mathbb{Z}_p \vee \Sigma^{-1} H\mathbb{Z}_p$$

Proof. There is a cofibration sequence

$$\mathrm{TC}(\mathbb{F}_p; p) \to \mathrm{TR}(\mathbb{F}_p, p) \xrightarrow{1-F} \mathrm{TR}(\mathbb{F}_p, p).$$

Note that

$$\pi_k(\mathrm{TR}(\mathbb{F}_p, p)) = \varprojlim_R \pi_k(\mathrm{THH}(\mathbb{F}_p)^{C_{p^n}}).$$

This implies $\pi_k(\mathrm{TR}(\mathbb{F}_p, p)) = \mathbb{Z}_p$ for $k = 0$ and is 0 otherwise.

Thus the cofibration sequence yields

$$\pi_0 \mathrm{TC}(\mathbb{F}_p, p) = \mathbb{Z}_p, \ \pi_{-1} \mathrm{TC}(\mathbb{F}_p, p) = \mathbb{Z}_p$$

Since $\mathrm{TC}(\mathbb{F}_p)$ is p-complete it is equal to $\mathrm{TC}(\mathbb{F}_p, p)$ by Goodwillie's theorem, and the result follows. □

4.1.3 TC for perfect fields over $\mathbb{F}_p$

Let us now state how to generalise the calculations to $k/\mathbb{F}_p$ perfect. In what follows we will use the ring of p-typical Witt vectors of k. We recommet the reader looks at [HM97, Section. 3] for an introduction.

Let $W(k)$ be the ring of p-typical Witt vectors of k. This ring comes with two (pertinent to us) endomorphisms, the Verschiebung maps V and the Frobenius, F.

The ring of Witt vectors of length n, $W_n(k)$is the quotient $W(k)/V^nW(k)$.

Theorem 4.10 (Witt). *If k is a perfect field of positive characteristic p then $W(k)$ is complete discrete valuation ring with residue field k and uniformizing element p. In particular $W(\mathbb{F}_p) = \mathbb{Z}_p$*

For a proof see [HM97, Thm. 3.1].

Now one can compute the integral homotopy groups of $\mathrm{THH}(k)^{C_{p^n}}$. The result is

Theorem 4.11. *For any perfect field k of positive characteristic p*

$$\pi_* \mathrm{THH}(k)^{C_{p^n}} \cong S_{W_{n+1}(k)}\{\sigma_n\}, \quad \deg \sigma_n = 2$$

and $F(\sigma_n) = \sigma_{n-1}, V(\sigma_{n-1}) = p\sigma_n$ and $R(\sigma_n) = p\lambda_n\sigma_{n-1}$ where $\lambda_n \in W_{n+1}(\mathbb{F}_p) = \mathbb{Z}/p^{n+1}$ is a unit.

Proof. See [HM97, Theorem. 5.5] □

Theorem 4.12. *If k a perfect field of characteristic p, then*

$$\mathrm{TC}(k) = H\mathbb{Z}_p \vee \Sigma^{-1}H(\mathrm{coker}(1-F))$$

Here F is the Frobenius $F : W(k) \to W(k)$ on Witt vectors.

Proof. The calculation in Theorem4.11 implies that $\mathrm{TR}(k) = HW(k)$ i.e. the Eilenberg Maclane spectra on $W(k)$.

There is a cofibration

$$\mathrm{TC}(k) \to \mathrm{TR}(k) \xrightarrow{1-F} \mathrm{TR}(k).$$

This gives an exact sequence

$$0 \to \pi_0 \mathrm{TC}(k) \to W(k) \xrightarrow{1-F} W(k) \to \pi_{-1}\mathrm{TC}(k) \to 0.$$

Now for $k = \mathbb{F}_p$ the Frobenius is the 0 endomorphism, and this affirms our previous result. Thus since $\mathrm{TC}(k)$ is a module over $\mathrm{TC}(\mathbb{F}_p)$ it is an Eilenberg-Maclane spectrum. Therefore we have

$$\mathrm{TC}(k) = H\mathbb{Z}_p \vee \Sigma^{-1}H(\mathrm{coker}(1-F))$$

□

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